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Machine learning-based in-process monitoring for laser deep penetration welding: A survey

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ABSTRACT

In-process monitoring (IPM) of laser deep penetration welding (LDPW) has witnessed a rapid growth in approaches that embrace machine learning algorithms, utilizing raw sensor input to generate various weld quality evaluations, instead of concentrating on thermomechanical modeling that is hypotheses-driven and hence biased by it. Benefitting from the capability to unravel hidden interactions in the complex laser welding process, numerous data-driven IPM methods have been proposed to address different problems in this area. In this survey, we present a comprehensive analysis of both classical and recent studies, covering the unique physical mechanisms, sensing techniques, methodologies, strengths, and limitations of machine learning-based IPM-LDPW systems. We delve into several critical tasks, including mechanical performance prediction, weld penetration estimation, and weld defects detection. Meanwhile, we explore the latest developments in deep learning and how to incorporate these techniques into IPM systems for LDPW. To inspire future research, we outline unresolved challenges and explore potential opportunities and new perspectives for addressing these challenges.

1. Introduction

Laser deep penetration welding (LDPW) is a fusion welding technique that leverages the motion of a laser-induced evaporated hole and the subsequent solidification of molten metal to join the workpieces (Yu et al., 2023). With its high aspect ratio (welding depth to seam width), narrow heat-affected zone, high processing speed, and relatively easy automation, LDPW is widely used in various industries, such as automotive, aerospace, medical, electronics, and polymers (Acherjee et al., 2011; Kumar et al., 2024; Huang et al., 2024). With the increasing applications related to human health and life, monitoring and controlling the weld quality has become even more important, especially in the production of electric vehicle batteries (Sadeghian and Iqbal, 2022).

Despite the different production scenarios, there are several common issues that obstacle the consistency of the weld quality, as illustrated in Fig. 1(a)–(c). The variety of welding application scenarios complex combinations of the joint types and the scan path, which in turn leads to challenges in the consistency and stability of the welding process. Furthermore, the formation of welds is primarily determined by the heat exchange area, which includes the molten pool and keyhole (Ye and Chen, 2002). Thus, it is intuitive and essential to correlate any physical phenomenon of this region that can be sensed with the weld quality. However, the region of laser-matter interaction is partially enclosed, and the dimensions and the timescales of the monitored fluid objects are on the order of millimeters and microseconds respectively (Fotovvati et al., 2018). These make it challenging to comprehensively capture the entire life cycle of the dynamic physical phenomenon. Besides the inherent limitations of the physical process, other factors encountered during the process also would lead to production failures, such as alteration of laser beam properties, contaminations on the workpiece, material microstructure defects (Stavridis et al., 2018), non-uniform in the material properties (Shevchik et al., 2019), and thermal distortion of the workpiece. These problems have posed challenges for modeling physical processes and assessing weld quality.

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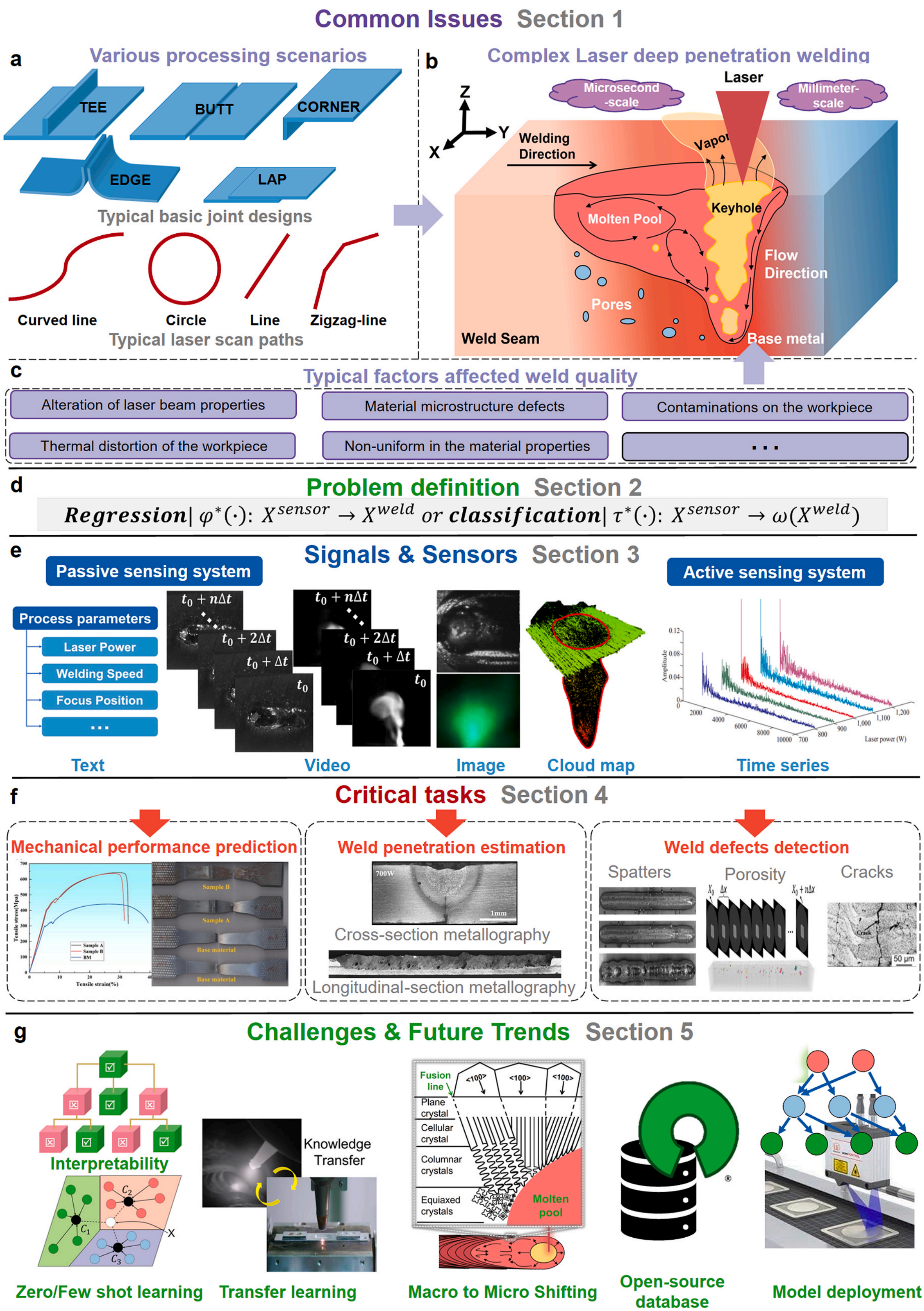

Fig. 1. A glance at this survey. I. Common issues for IPM-LDPW encountered during production. II. Problem definition of ML-based IPM-LDPW. III. Data types of signal acquisition settings, including text, video (Zhang et al., 2018a), image, cloud map (Ma et al., 2022a), and time series (Zhang et al., 2018a). IV. Three representative tasks in ML-based IPM-LDPW, cover mechanical performance prediction (Hu et al., 2024), weld penetration estimation, and weld defect detection (D. You et al., 2014a; Mondal et al., 2022). V. Critical challenges and corresponding future opportunities.

In light of this, to keep production results within deterministic boundaries, monitoring and modeling the variations during welding is vital and fundamental for quality control (Kaierle, 2008). Although thermomechanical and multi-physical simulations have shown remarkable advancements in dynamic modeling and mechanisms, the hypotheses-driven bias could eventually lead to unacceptable discrepancies between estimated and actual dynamics (Lorenz, 2004). As the rapid development of process signals acquisition technologies, machine learning on in-process monitoring (IPM) and modeling of weld quality has been attracting more and more attention, especially deep learning in recent five years (Cai et al., 2019, 2020; Wu et al., 2022; Yu et al., 2023). The data-driven learning methods have exhibited superior generalization abilities in utilizing knowledge-based features and demonstrated significant potential in extracting meaningful representations of underlying patterns in diverse data types, such as texts, videos, images, cloud points, and time series obtained from LDPW. However, machine learning (particularly deep learning) on in-process monitoring of laser deep penetration welding (IPM-LDPW) still faces several significant challenges (B. Wang et al., 2020; Yu et al., 2023), such as the interpretability, multimodal learning, small-scale datasets, the model deployment in a production line. On this basis, this work focuses on the analysis of both classical machine learning methods and recent deep learning methods that have been used in IPM-LDPW, and most of these methods belong to supervised learning.

As an important research area, many surveys of machine learning or deep learning on welding IPM are also available. Table 1 enlists some recent surveys on machine learning-based in-process monitoring of welding (ML-based IPM-W), which focus on different topics. According to the review by (Zhang et al., 2020), laser welding has complex process characteristics that require multiple sensing capabilities. Compared to these reviews (B. Wang et al., 2020; Y. M. Zhang et al., 2020; Cheng et al., 2021; Madhvacharyula et al., 2022; Z. Wang et al., 2023; Yu et al., 2023; Amarnath et al., 2023; Guo et al., 2024) that do not solely focus on laser welding, this work thoroughly explains the unique dynamic phenomena and elucidates the corresponding monitoring mechanism during laser welding. Compared with the existing literature (Cai et al., 2020; Wu et al., 2022) on IPM-LDPW, the major contributions of this work can be summarized as follows:

1) To the best of our knowledge, this is the first survey that summarizes the procedures on ML-based IPM-LDPW, formally defines the task of the ML-based IPM-LDPW, and outlines the common sensors, signals, and the typical data acquisition settings.
2) This paper presents the latest advancements in deep learning for IPM-LDPW, providing readers with state-of-the-art methods.
3) For three representative monitoring tasks, comprehensive summaries of encountered problems and intuitive explanations of why certain model inputs work are provided.

The structure of this survey is illustrated in Fig. 1(a)–(g). Section 1 introduces the common issues for IPM-LDPW encountered during production and the motivation for surveying the works on ML-based IPM-LDPW. Section 2 summarizes the common procedures of ML-based IPM-LDPW and formally defines the task of the ML-based IPM-LDPW. Section 3 provides an overview of the evolution of sensor technology corresponding to welding phenomena and the typical data acquisition settings. Section 4 reviews the ML-based methods for mechanical performance prediction, weld penetration estimation, and weld defects detection. Section 5 summarizes a series of critical challenges and corresponding future opportunities. Section 6 concludes this paper.

2. Machine learning-based in-process monitoring of laser deep penetration welding

Intelligent welding involves four functional subsystems: monitoring, diagnosis and understanding, evaluation and prediction, and control (B.

Table 1
Summary of ML-based IPM-W surveys.

Article Name	Welding technique scope	Description related to ML-based IPM-W
Intelligent welding system technologies: State-of-the-art review and perspectives (B. Wang et al., 2020)	Unspecified	This review provides an overview of sensors, signal processing techniques, feature engineering, and ML-based monitoring with a primary focus on traditional ML methods during different welding.
Application of sensing techniques and artificial intelligence-based methods to laser welding real-time monitoring: A critical review of recent literature (Cai et al., 2020)	Laser welding	This study summarizes physical phenomena related to laser welding, sensing techniques, and traditional ML-based monitoring techniques (mostly traditional ML) during laser welding.
Advanced Welding Manufacturing: A Brief Analysis and Review of Challenges and Solutions (Y. M. Zhang et al., 2020)	GTAW, GMAW, PAW, K-TIG, LDPW, FSW, AC-GMAW and double-electrode GMAW	This work presents an overview of common sensing techniques used during welding and ML-based modeling techniques for estimating penetration during diverse fusion welding.
Real-time sensing of gas metal arc welding process – A literature review and analysis (Cheng et al., 2021)	GMAW	This paper focuses on monitoring weld pools during GMAW using computer vision and deep learning techniques.
Progress and perspectives of in-situ optical monitoring in laser beam welding: Sensing, characterization and modeling (Wu et al., 2022)	LDPW	This review provides an overview of optical sensing, signal processing, feature engineering, ML for penetration estimation, and detection of weld defects during LDPW.
In situ detection of welding defects: a review (Madhvacharyula et al., 2022)	Unspecified	This study provides an overview of non-destructive testing techniques for monitoring welding defects using signals from acoustics, optics, welding current, and voltage, with an emphasis on classification algorithms.
Automatic detection of defects in welding using deep learning - a systematic review (Amarnath et al., 2023)	TIG	This review discusses the use of convolutional neural networks for monitoring joint penetration with molten pool images during TIG and presents some lightweight models for IPM.
Deep learning based real-time and in-situ monitoring of weld penetration: Where we are and what are needed revolutionary solutions? (Yu et al., 2023)	Unspecified	This paper provides a comprehensive overview of machine learning and deep learning techniques used to monitor weld penetration during various welding processes, with a focus on using image data as the input for the model.
Penetration recognition based on machine learning in arc welding: a review (Z. Wang et al., 2023)	Arc welding	This study provides an overview of machine learning techniques used for recognizing

(continued on next page)

Table 1 (*continued*)

Article Name	Welding technique scope	Description related to ML-based IPM-W
		penetration using weld pool images, arc sound, and arc voltage during arc welding. Deep learning methods are also introduced in detail.
Progress, challenges and trends on vision sensing technologies in automatic/ intelligent robotic welding: State-of-the-art review (Guo et al., 2024)	Automatic/intelligent robotic welding	This literature summarizes welding defect detection technology based on infrared sensing and visible light sensing by comparing ML-based intelligent detection algorithms and traditional detection algorithms.

'GTAW' represents Gas tungsten arc welding, 'GMAW' represents gas metal arc welding, 'PAW' represents keyhole plasma arc welding, 'K-TIG represents keyhole-tungsten inert gas, 'FSW' represents friction stir welding, 'AC-GMAW' represents alternating current GMAW.

Wang et al., 2020). As a typical intelligent welding technique, ML-based IPM-LDPW contains all the procedures of the first three functional systems except the control aspect in this paper. This section provides an overview of the procedures of the ML-based IPM-LDPW system, and abstracts and defines the most common problems encountered in ML-based IPM-LDPW.

2.1. Common procedures

Through a survey of more than 200 papers, we have summarized common ML-based paradigms for IPM-LDPW. Fig. 2 illustrates a schematic diagram of the common procedures for ML-based IPM-LDPW systems. In general, the modeling methods can be categorized into two paradigms based on the requirement of feature engineering, namely traditional modeling and end-to-end learning. As a result, the ML-based IPM-LDPW pipeline can be outlined in either a 5-step or 4-step configuration.

Through sensors that interface with the LDPW process, physical phenomenon data is gathered and then preprocessed to remove inconsistencies or errors. Since most of the work is supervised or semi-supervised, data labeling procedures are crucial for modeling based on data-driven learning. Usually, after welding experiments, the weld seam undergoes cross-sectional cutting, mounting, standard mechanical polishing, etching, and optical microscopy observation to obtain macro features for outputting. The labeling procedure can be time-consuming and costly. Nowadays, non-destructive methods, such as Synchrotron X-ray imaging, have also been employed for data labeling, providing specific outputs (Paulson et al., 2020; Aminzadeh et al., 2024).

In the process of traditional ML-based modeling, data featurization plays a crucial role. This process involves extracting or creating meaningful features using either domain knowledge or model-validation experiments. These features are then scaled appropriately and the dimensionality of the dataset is reduced to train a better model. In the field of ML-based IPM-LDPW, physical features such as the perimeter, area, and temperature distribution of the molten pool and keyhole, the center of mass, area, and spectral intensity of the plasma plume, as well as the corresponding frequency-domain transformed features are commonly used for different quality evaluation tasks. However, when using end-to-end learning architectures, deep learning can automate feature engineering by using a large amount of labeled data.

After preparing the data, learning algorithms are selected carefully and trained to predict various criteria for weld quality, such as mechanical properties, weld penetration, or weld defects. Cross-validation is commonly used during the training process to balance the bias-variation tradeoff to prevent overfitting or underfitting. Finally, a properly chosen and well-trained model is deployed to provide real-time monitoring and feedback for the actual production. Additionally, incremental learning can also be adopted to improve the robustness and reliability of this system, fine-tuned by data collected from the specific production line to achieve transfer learning from the laboratory environment to the actual production line.

2.2. Problem definition

We formally define the tasks of ML-based IPM-LDPW, as follows. Let $X_t \in R^D$ be the physical phenomena collected from the t-th moment during LDPW of the workpiece $X = \{X_1, X_2, \cdots, X_{t-n}, X_{t-n+1}, \cdots, X_n\}$. The target is to predict the physical phenomena of weld formation $X^{weld} = \{X_1^{weld}, X_2^{weld}, \cdots, X_{t-n}^{weld}, X_{t-n+1}^{weld}, \cdots, X_n^{weld}\}$ with $X_i^{weld} \in R^F (F \ll D)$ from the sensible physical phenomena $X^{sen} = \{X_1^{sen}, X_2^{sen}, \cdots, X_{t-n}^{sen}, X_{t-n+1}^{sen}, \cdots, X_n^{sen}\}$ with $X_i^{sen} \in R^C (C \ll D)$. In an ideal situation, the ML-based IPM-LDPW aims at learning a feature representation mapping function, in regression form $\phi(\bullet) : X^{sen} \rightarrow X^{weld}$ or a classification form $\tau(\bullet) : X^{sen} \rightarrow \omega(X^{weld})$ where $\omega(\bullet)$ denotes the score function.

To some extent, the sensing space is a subset of the space of weld formation, as nearly all surface physical phenomena are part of the weld formation. Hence, the goal of the ϕ and τ functions is to infer the higher-dimension information of weld formation by its subspace information. In other words, deducing solid-liquid boundaries and solidification laws of weld pool from its perceptible surface features. Furthermore, as mentioned in Fig. 1(a)–(c), the area where the laser interacts with the material is only partially open. This causes the sensors to not be able to fully capture the internal physical phenomena of LDPW. Thus, the dimension of sensing space is less than that of the physical phenomena of LDPW.

Besides that, the limitation of the adopted sensors not only compresses and transforms the information dimension: $f(\bullet) : X^{sen} \rightarrow X^{sen\_low}$, but also adds noise: $X^{sensor} = g(X^{sen\_low}, \alpha, \varepsilon)$. Here, $X^{sen\_low} \in R^N$ (where $N < C$) denotes the dimension-compressed sensible physical phenomena, while $X^{sensor} \in R^M$ (where $M < N$) represents the data converted from the X^{sen_low} by the adopted sensors. Additionally, $\alpha \in R^{t \times s}$ refers to the insensible factors that influence the X^{sen_low}, and ε refer to noise collected during welding, respectively. Therefore, in real practice, the ML-based IPM-LDPW aims at learning a feature representation mapping function, in either regression form $\phi^*(\bullet) : X^{sensor} \rightarrow X^{weld}$ or classification form $\tau^*(\bullet) : X^{sensor} \rightarrow \omega(X^{weld})$.

In general, constructing the transfer function g can be difficult

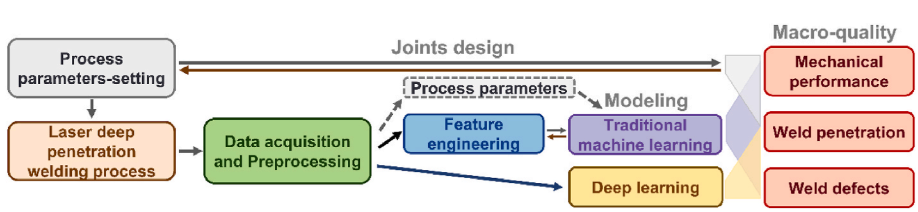

Fig. 2. Common procedures of ML-based IPM-LDPW.

because the α is nonlinear, time-varying, and hard to sense. This includes, but is not limited to changes in laser beam properties, contamination on the workpiece, material microstructure defects, non-uniform material properties, and thermal distortion of the workpiece. However, by learning the underlying patterns from hand-crafted or automated feature space, the data collected by the sensors: X^{sensor} can respond to patterns of change induced by the α. This can help to reduce the bias and variance introduced by the dimensionality reduction and noise. Despite this, the noise ε sets the upper limit of data-driven learning performance, which is determined by the capabilities of the adopted sensors.

3. Data acquisition

Observational data produced during the on-the-fly monitoring of LDPW has been a common practice to correlate the weld quality in this field. These data ought to reflect the underlying physical principles that govern their generation, and, in principle, can be used as a weak mechanism for embedding these principles into an ML model during its training phase (Karniadakis et al., 2021). Therefore, it is important to have a well-designed data acquisition system that accurately captures physical principles without aliasing to ensure high-quality data for data-driven learning.

3.1. Sensors and signals

There are two primary data source objects that are monitored during LDPW: the plasma plume, and the molten pool and keyhole. Depending on which of these objects is being focused on, the sensors used can be generally classified into three categories: optical emissions, acoustic emissions, and charge potential. Table 2 lists the common sensors for monitoring signals sourced from the above objects.

3.1.1. Optical emissions signal

Optical Emissions (OE) signals from the welding zone are the most commonly used in the IPM-LDPW, and they are derived from three sources: 1) reflected laser light, 2) thermal radiation from the weld zone, and 3) radiation intensity of the plasma (Olsson et al., 2011).

Reflected laser light varies its intensity depending on the surface morphology of the molten pool, and extreme variation can lead to violent fluctuations in reflected light intensity. Photodiodes with a narrow bandpass around the laser wavelength are the most commonly used sensor for monitoring it. Since reflected laser light signal has low correlations with plasma or thermal signals and have precision limits (Olsson et al., 2011; You et al., 2015; Shevchik et al., 2019), they are generally used as auxiliary signals. However, this signal is useful in detecting excessive contamination of protective lenses in actual production.

Thermal radiation from the weld zone reflects the temperature gradient in the welding area and is commonly collected by infrared sensors sensitive to wavelengths from 1300 to 1800 nm, such as CCD- or CMOS cameras (Dorsch et al., 2012), pyrometer (Xiao et al., 2020a), near-infrared (NIR) camera (Dorsch et al., 2013), and photodiode (D. You et al., 2014a). The radiation intensity of the plasma is mainly in the ultraviolet (UV) range of the spectrum, and it is used to measure the stability of the welding process (Xiao et al., 2016). Photodiode, spectrometer, CCD- or CMOS- camera coupled with UV bandpass filter are the frequently adopted sensors for OE of plasma plume (D. You et al., 2014a; You et al., 2015; Deng et al., 2020). Currently, research focused on the monitoring and diagnosis of LDPW has established connections between weld defects, such as blowout (Olsson et al., 2011), oxidation, lack of penetration (Sibillano et al., 2007), porosity (Harooni et al., 2014a), humping (Xiao et al., 2020a; You et al., 2015), and burn-through (You et al., 2015), with OE signals.

Although OE signaling is widely used in industrial applications due to its contactless benefits, it is limited by the trade-off between reliability and time consumption. Photodiodes, pyrometers, or

Table 2
Introduction of common sensors.

Sensor	Signals	Pros	Cons
Photodiodes	Reflected laser light, thermal radiation from the weld zone, Radiation intensity of the plasma	Signals are sampled at a high rate with computational efficiency, and can detect protective lens contamination, molten pool variation, energy absorption, and molten spray capture.	Signals have low precision, lack spatial dimension information, and cannot detect internal weld defects.
Camera	Thermal radiation from the weld zone, radiation intensity of the plasma	Sensors can capture both the temperature distribution and spatiotemporal information of molten pool and keyhole, as well as changes in the plasma plume's 2D or 3D morphology over time.	Sensors pose challenges in real-time frame transmitting with high resolution, and dense images lead to difficulties in feature learning and low computational efficiency.
Pyrometer	Thermal radiation from the weld zone	Sensors measure the temperature with high accuracy, high sampling rate, and high computational efficiency.	Signals can be influenced by the vaporization plume, as well as the calibration of the object's emissivity.
Spectrometer	Radiation intensity of the plasma	Sensors can provide information on chemical composition, focal length, and thermal effects during laser processing.	Sensors are hard to setup, work short distances, high operating and maintenance costs.
Microphone	Acoustic emissions with a band range of 20 Hz to 20 kHz	Signals can indicate the quality of the weld and the stability of the welding process	Sensors in industrial settings suffer from noise interference and acoustic wave transmission delays.
Piezoelectric transducer/ resonant sensor	Acoustic emissions within a band range of less than 200 kHz	Signals can detect volumetric variations within the laser-material interaction area, cracks and pores.	Sensors must be in direct contact with the object being measured, limiting its flexibility.
Copper probe	Electrical signal that arises due to the plasma sheath effect in laser-induced plasma	Sensors can measure plasma temperature with rapid response time and cost-effectiveness.	Sensors require physical contact with plasma limited flexibility and have modest precision in measuring plasma.
Visual (VIS) camera combined with an auxiliary illuminant system	The light with a selected wavelength of the auxiliary illuminant reflected from the molten pool and keyhole	Sensors can capture the molten pool and keyhole without interference from metal vapors.	Sensors' high precision and sampling rate come at a relatively high cost, while signals suffer from low computational efficiency and incur high

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Table 2 (*continued*)

Sensor	Signals	Pros	Cons
			computational costs.
Optical coherence tomography	Spectral interference of reference light beam and measuring light beam.	Sensors can measure the keyhole depth variation in real-time and high accuracy.	Signals failure for unstable keyhole, and sensors are very expensive.
Electromagnetic acoustic transducer	The ultrasonic waves propagating through a molten pool (liquid phase), a solid phase, and only the solid phase near the molten pool.	Sensors can detect the solid-liquid interface of molten pool in high accuracy.	Sensors requires complex setup imposes significant limitations in flexibility
X-ray tomography	Intensity of x-rays after passing through matter.	Sensors have high precision and high speed.	Signals can be harmful to human health and cannot penetrate thick sheets, sensors have limited flexibility.

spectrometer-based sensing are commonly used to measure the average value of local light emission intensity. However, several coincident process instabilities also change the light emission intensity, including turbulences, surface ripples, plume transience, and the formation and collapse of the keyhole. These dynamic phenomena can be challenging to discern with point sensors, which can lead to difficulty in diagnosing the presence of macrostructural defects such as porosity and cracks (Vakili-Farahani et al., 2017).

Photodiodes offer advantages in terms of sampling, computational speed, affordability, and simplicity, making them suitable for process monitoring and controlling in production. However, they may not accurately capture temperature variations since the emissivity of the thermal source is not a constant (Xiao et al., 2020a). Spectrometers are sensitive to the dynamic behavior of the plasma but are not suitable for scenarios that suppress the plasma shielding effect. The accuracy of pyrometers is affected by the intense interference from the plasma plume inside the keyhole.

Camera-based vision sensing and thermographic imaging are commonly used to capture information in 2D, 3D, 4D, or even 5D dimensions carried by images and videos. Through continuous sequences of images, they can capture spatial (1-3D), temporal, and temperature information in the region of laser-material interaction, facilitating the detection of intricate weld defects such as porosity and cracks. However, the processing of dense matrices also imposes notable time constraints, limiting its application in industrial production.

3.1.2. Acoustic emissions signals

Acoustic Emissions (AE) signals can be categorized into two types - structure-borne acoustic and air-borne acoustic emission signals. Structure-borne AE arises from elastic stress waves that occur due to alternating pressures of melted material against a solid surface. It primarily originates from sources like cracks, pores, phase transformations, and back-reflected laser energy. On the other hand, air-borne AE results from pressure waves that occur due to the recoil force and thermal vibration of metallic vapor. These waves can reflect phenomena such as the formation of a plasma plume, keyhole oscillation, and molten pool variation, which are influenced by temperature fluctuations and mass transfer within the molten pool (Dixon and Lewis, 1985; Gu and Duley, 1996; Li, 2002; Luo et al., 2019; Shevchik et al., 2019).

AE sensors can be categorized into contact and non-contact types. Contact sensors, such as piezoelectric or resonant sensors, directly interface with the surface of the workpiece to collect structure-borne AE signals. Meanwhile, non-contact sensors, like microphones, acquire air-borne AE signals without the need for direct contact with the surface of the workpiece. Contact AE detection predominantly monitors stress waves, typically within a band range of less than 200 kHz (Fang et al., 1997). In contrast, non-contact AE detection focuses on pressure waves, usually within a monitored band range of 20 Hz to 20 kHz (Huang and Kovacevic, 2009a).

Unlike OE signals gathered from the outer surface of the welding zone, AE signals can capture volumetric variations within the laser-material interaction area. Studies have demonstrated the correlation between AE signals and weld defects, particularly cracks (Fang et al., 1997), pores (Sun et al., 2001), and weld penetration (Cao et al., 2023; Luo et al., 2023). Additionally, research has explored AE characteristics during the formation of welds (Farson et al., 1998) and plasma (Conesa et al., 2004), as well as the relationship between AE and temperature (Li, 2002).

While AE signal monitoring holds significant potential in pulse laser welding monitoring, research in this area has seen a decline in recent years, largely due to practical constraints in industrial applications. In the case of structural emission monitoring, the necessity for contact-mounted sensors renders it unsuitable for mass production. Airborne emission detection often encounters difficulties in obtaining meaningful signals in challenging and noisy industrial settings. Moreover, the inherent delay in the transmission of acoustic waves through air poses a limitation for its application in real-time monitoring and control.

3.1.3. Charge potential signal

Charge Potential (CP) signal is an electrical signal that arises due to the plasma sheath effect in laser-induced plasma. The plasma is composed of positively charged ions and negatively charged electrons. This results in the accumulation of negative charges on the metal surface due to the movement of these particles. The negative potential thus created hinders the flow of electrons toward the metal surface while facilitating the movement of positive ions toward it. The current densities of positive ions and electrons reach an equilibrium, leading to the formation of the plasma sheath. To gather CP signals, a circuit sensor is used. One end of this sensor is connected to the substrate, while the other end is connected to a probe located either in the plasma region or the welding nozzle, as shown in Fig. 3 (Huang et al., 2019a).

(Zhao et al., 2017) noted that distinct dominant frequency peaks in the CP signal spectrum indicate the presence of a keyhole. Furthermore, with an increase in penetration depth, the characteristic frequency peaks decrease from approximately 1150 Hz–400 Hz, ultimately reducing to one peak when full penetration is attained. This method has proven effective in measuring weld penetration, welding performance, and plasma temperature (Huang et al., 2019a; Li et al., 1996).

While CP measurement, as a contact-based sensor, offers the benefits of rapid response time and cost-effectiveness, its measurement accuracy is relatively modest, which can present challenges when precise evaluation is necessary. Additionally, it may encounter limitations in its application to processing equipment like remote laser welding, primarily due to circuit settings and installation requirements.

3.1.4. Molten pool and keyhole geometry

Geometric measurement of the molten pool and keyhole during the welding process is challenging due to the photothermal and vaporization phenomena. The weak reflected light and intense thermal radiation from the welding zone make it difficult to obtain a clear picture of the molten pool and keyhole using camera-based vision, as shown in Fig. 4 (a) (Yi Zhang et al., 2019). Researchers have attempted to address this issue using various methods.

The illuminant system is one of the effective and feasible solutions, which strengthens the weak illumination of the molten pool using a laser light source and attenuates the thermal radiations from the keyhole using optical filters, as shown in Fig. 4 (b) (Yi Zhang et al., 2019). This

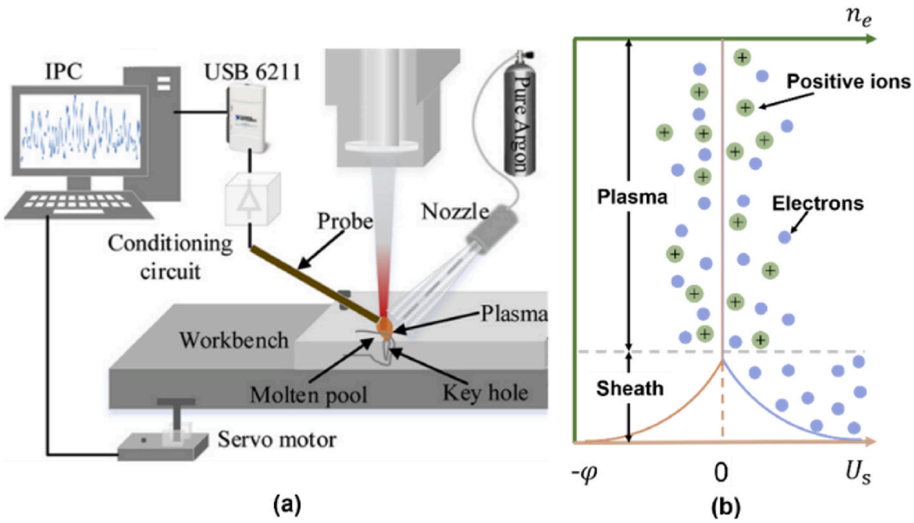


Fig. 3. Schematic diagram of measuring principle for CP signals, (a) the circuit used to collect CP signals, (b) plasma sheath measuring principle (Huang et al., 2019a).

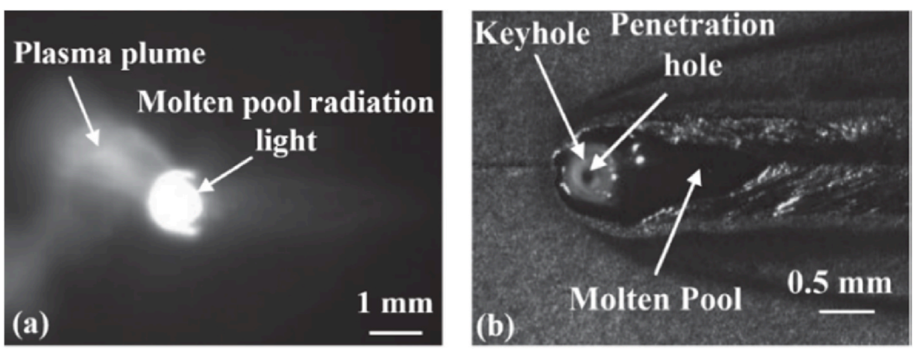


Fig. 4. Images of the molten pool and keyhole by (a) without an illuminant system and (b) with an illuminant system (Yi Zhang et al., 2019). More geometric information was enhanced by an auxiliary illuminant system.

deliberate hardware selection eliminates the need for introducing computational time for noise reduction algorithms directly at the source. Furthermore, the dynamic changes in ripples and the keyhole inlet on the molten pool surface offer the potential to uncover intricate welding defects. However, the camera cost, which is positively correlated with the sampling frequency, and the redundancy time of matrix operations remain the most significant drawbacks of the scheme.

Optical coherence tomography (OCT), also known as inline coherent imaging or low coherence interferometry, has been implemented in a production environment for geometry measurement of the molten pool and keyhole (Dupriez and Denkl, 2017). OCT exhibits the ability to rapidly measure the etch rate and stochastic melt relaxation, without being blinded by intense machining light or incoherent plasma emissions (Webster et al., 2011). The increased melt depth resulting from convective heat transfer within the melt pool is the primary factor contributing to the accuracy of OCT. Additionally, it's important to note that OCT technology is constrained by its detection principle and may not be suitable for measuring J-shaped keyholes. Moreover, the high cost of acquisition serves as a significant impediment to its widespread implementation in massive industrial production.

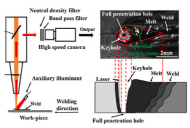
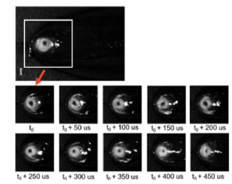
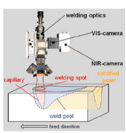
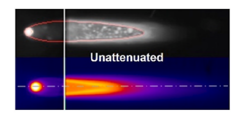
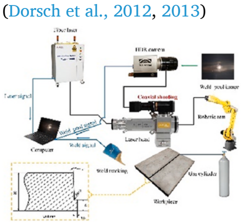
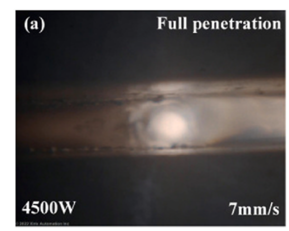
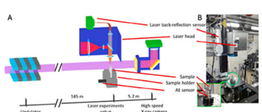
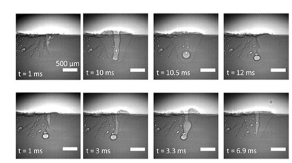
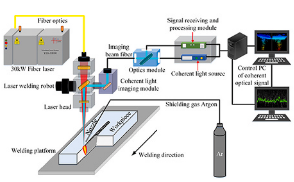
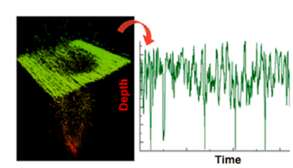
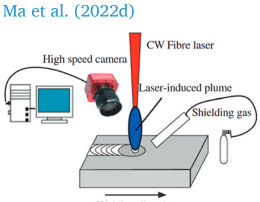
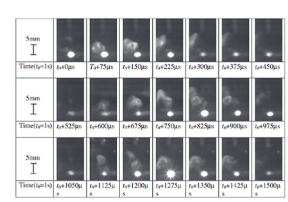
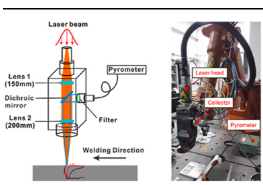
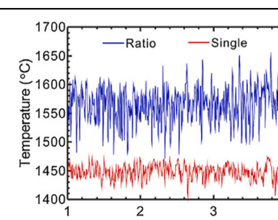
Electromagnetic acoustic transducer (EMAT) represents an alternative measuring method that relies on external excitation. It enables the measurement of the geometry of the molten pool and keyhole, determined by the solid-liquid interface, with an error margin of ± 0.5 mm through the use of electromagnetic coupling to excite and receive ultrasonic waves (Mizota et al., 2015). However, it is worth noting that while EMAT offers relatively highly accurate measurements of the melt pool, its complex setup imposes significant limitations on the range of parts it can be effectively used for. Additionally, the method necessitates the introduction of additional clamping devices.

The X-ray-based system monitors the weld area by the extent to which X-rays are absorbed and has demonstrated the ability to monitor the internal structure of the weld, such as the variation of weld geometry, porosity evolution (Hojjatzadeh et al., 2020) and even the residual stress distribution (Kong and Kovacevic, 2010). X-ray tomography offers the most precise measurements of the molten pool and keyhole. However, its suitability for thick sheets is constrained by the synchrotron light source, and it necessitates a radiation-shielded room to safeguard the well-being of personnel. These drawbacks have hindered its transition from the laboratory to the industrial production setting.

3.2. Typical acquisition methods

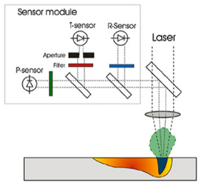
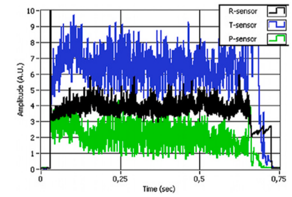
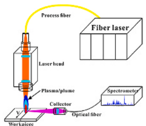
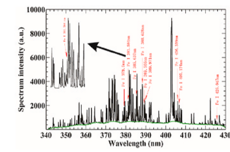
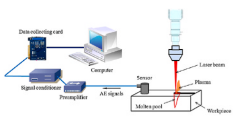
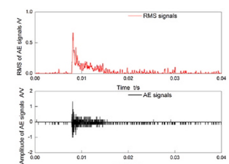
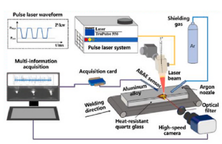
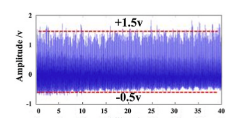
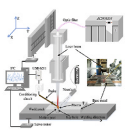
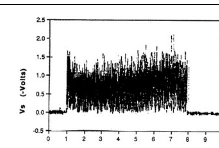
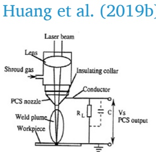
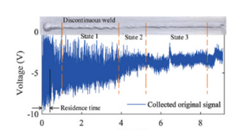
Sect.3.2 provides the acquisition setup of laser welding monitoring from most of the current researches (Wu et al., 2022; Aminzadeh et al., 2023; Cao et al., 2023; Guo et al., 2024). The data available gained by the acquisition setup is continually improved in spatial-temporal-thermal resolution. It is also expanding to include complex data types, such as images, videos, multivariate time series, audios, point clouds and so on. Common data acquisition settings for non-destructive in-process monitoring of laser welding are listed in Table 3. All of these acquisition methods can monitor the quality of welds in laboratory conditions. However, not all of them are used for quality inspection in real-world production environments due to time and cost constraints. Specifically, data acquisition systems using photodiodes are the most common sensors adopted for welding anomaly detection for their fast response, low data dimensions, and low costs. Coherent optical system is advantageous for measuring the depth

Table 3
Summary of typical data acquisition settings for LDPW monitoring.

Schematic Diagram	Setting Specifications	Data Formats
	Geometry of Plasma Plume, and Molten pool and Keyhole	
 Lu et al. (2020a)	Visible camera: High-speed camera (Photrom Fastcam SA4), neutral density filter (NDFI5003, transmittance: 50%), bandpass filter (central wavelength: 808 nm), semiconductor laser illumination system (wavelength: 808 nm)	
 (Dorsch et al., 2012, 2013)	Infrared (IR) camera: InGaAs-camera (sensitive: 900 nm–1700 nm), neutral density filter (transmittance: 10%)	
 Wu et al. (2024)	High-dynamic-range (HDR) camera: WJ-1100 Xiris HDR camera (max fps: 55, resolution: 1280*1024, HDR>140 dB)	
 Shevchik et al. (2019)	X-ray tomography: Imaging beamline ID19 (European Synchrotron)	
 Ma et al. (2022d)	Optical coherent imaging: IPG-LDD-700 coherent optical system	 Ma et al. (2022a)
 Wang et al. (2012)	Visible camera: High-speed camera (Photonfocus)	
	Light Radiation from Plasma Plume, Molten pool and Keyhole	
 Xiao et al. (2020b)	Pyrometer (range of temperature: 600–2500 °C)	

(continued on next page)

Table 3 (*continued*)

Schematic Diagram	Setting Specifications	Data Formats
 Eriksson et al. (2010)	IR/UV (ultraviolet)/VIS photodiodes	
 Xiao et al. (2016)	Spectrometer (Ocean Optics, spectral window: 340–430 nm)	
	Acoustic emissions of welding zone	
 Luo et al. (2019)	Structure-borne acoustic emission sensor (no specifications provided)	
 Luo et al. (2023)	Air-borne acoustic emission sensor (BSWA TECH, Beijing, China, MA231 preamplifier)	
	Charge potential of Plasma	
 Huang et al. (2019b)	Copper probe, Data acquisition card (National Instruments, USB-6211)	
 Li et al. (1996)	Plasma charge sensor	

variation of keyhole during welding and provides high accuracy for stable welding processes. Visual cameras (not high-speed camera) mostly serve for localization and weld seam tracking.

4. Typical tasks in ML-based IPM-LDPW

Fig. 1(b) illustrates that laser welding is a complex physical system that involves fluid flow, and heat and mass transfer. Sensor networks have been used to observe and monitor the evolution of complex phenomena across several spatial and temporal scales with variable physical fidelity. Supervised models are commonly used to build a correlation between the observational data and the weld quality, where mechanical performance is the most common metric used to quantify weld quality.

Mechanical performance is not only a metric for weld quality but also an essential requirement in welding procedure specifications. Generally, mechanical performance is determined by the geometry of the weld and the presence of voids inside it, when the heat dissipation of the molten pool is invariant. As no sensor can comprehensively monitor the temperature profile of the molten pool, the mechanical performance of welds also can be indirectly evaluated by the weld penetration and defects. Therefore, the tasks of weld quality evaluation are divided into mechanical performance prediction, weld penetration estimation, and weld defect detection.

4.1. Mechanical performance prediction

In this task, machine learning models take the role of a "black box" to correlate the inputs with mechanical performance. Based on prior knowledge of laser energy absorption, most of the research focuses on predicting the mechanical performance using process parameters, which aids in joint design. Others rely on OE signals, as illustrated in Fig. 2.

4.1.1. Data preparation and analysis techniques

Creating a dataset to predict mechanical performance in ML-based

IPM-LDPW is a costly and time-consuming process that requires conducting numerous destructive experiments to obtain the labels of data. Therefore, covering a more complete sample space with the least amount of data is the target in this field. As a result, it is common practice to identify the process window through association analysis (Benyounis and Olabi, 2008). The most commonly used experimental design techniques to determine optimal welding process parameters include factorial design (Cicală et al., 2005; Escribano-García et al., 2022), Taguchi method (Benyounis and Olabi, 2008; Prabakaran and Kannan, 2019), response surface methodology (RSM) (Kumar et al., 2024) and Box-Behnken design (Benyounis et al., 2005; Olabi et al., 2013).

4.1.2. Process parameters-driven modeling

Mechanical properties are influenced by the metastable microstructures formed during the solidification of the molten pool. The thermal gradient emerges as the pivotal factor determining the microstructure of the weld (Kou, 2003), which is directly affected by the welding parameters. Hence, it is intuitively feasible to directly forecast the mechanical properties of welds by using in-process welding parameters. The use of process parameters for the prediction of mechanical performance based on the assumption that the process parameters are constant during the welding process can also be considered as IPM. It also can be used as a validation method to assess the abundance of anomaly data in the welding process.

In this context of modeling, linear regression (Kim and Park, 2011), polynomial regression (PR) (Lei et al., 2018), support vector regression (SVR) (Petković, 2017; Zhang and Zhou, 2019), RSM (Long et al., 2018; Yaakob et al., 2019; Zhao et al., 2012), grey relational analysis (Lin, 2012; Tamrin et al., 2014; Shrivastava and S. G. B, 2016), and Artificial Neural Networks (ANN) (Lin and Chou, 2008; Acherjee et al., 2011; Zhang et al., 2017; Kumar et al., 2024). Among these modeling methods, ANN is the most favored modeling method in this field, and global search algorithms such as GA (Sathiya et al., 2012; Yang et al., 2018) are preferred for locating the optimal space of process parameters.

The main challenges come from the inductive bias of the model inputs designing and model selections. Different laser welding techniques, such as laser oscillation welding (or laser transmission welding using incremental scanning technique), laser wire-fill welding, and laser-MIG hybrid welding, along with the distinct material properties of the workpiece, such as dissimilar materials, make the physical and chemical phenomena within the molten pool complex. These complex transformations pose a significant challenge not only for numerical simulation efforts but also for machine learning modeling. Table 4 lists the various scenarios tackled by process parameters-driven ANN. Particularly, (Acherjee et al., 2011; Kumar et al., 2024) addressed the issue of induction bias caused by different welding techniques. They predicted the mechanical properties of polymer joints with different properties by applying the factors that affect the laser energy input during welding. Both studies focused on searching for effective model architectures and found that deeper artificial neural networks were better at identifying correlations between heterogeneous materials. Furthermore, it can be concluded that the use of multi-task learning with shared weights has shown promising outcomes in determining the joint distribution of geometric and mechanical property characteristics of welded joints. This approach enhances the interpretability and performance of the model effectively. In addition, well-validated multi-objective optimization algorithms, such as the non-dominated sorted genetic algorithm (Kumar et al., 2024), are very important to avoid getting stuck at the local optima.

The inverse solution of the mechanical performance prediction task, which enables the optimization of welding parameters, is another common application of ANN. (Thi Tien et al., 2023) developed three surrogate models based on the simulation trained ANN to predict the melt pool geometry and copper concentration for all conceivable parameter combinations in the design space, as illustrated in Fig. 5. The processing maps of producing dissimilar joints were constructed to identify the optimal parameters for achieving no cracks or pores in the weld and enhancing the mechanical and electrical properties of joints.

4.1.3. Process signals-based methods

Fluctuations induced by the external environment cannot be reflected by the characteristics of welding parameters. Therefore, it is not feasible to assess the dynamics of the laser-material interaction area

Table 4
ANNs for tackling different challenges in LDPW monitoring.

Scenarios	Input	Network architecture	Imp.	Output	Ref.
Laser transmission welding of natural and opaque (containing 0.2% wt. carbon black) acrylics plaques	Power, welding speed, stand-off-distance, clamp pressure	4-4-5-2	Back-propagation (BP) algorithm	Lap-shear strength, weld-seam width	Acherjee et al. (2011)
Laser welding ofsuper austenitic stainless steel (AISI 904 L)	Beam power, travel speed, focal position	3-100-10-3	Genetic algorithm (GA)	Tensile strength, penetration depth, bead with	Sathiya et al. (2012)
Laser welding of NiTinol	Welding speed, shielding gas blown distance, focal position, laser power	4-8-1	Legvenberg-Marquardt, GA	Penetration depth, bead width, hardness, corrosion current density	Kannan et al. (2017)
Laser welding of low carbon steel (Q235) and stainless steel (SUS301L-HT)	Laser power, welding speed, focal position, gap	4-10-3	GA,particle swarm optimization (PSO)	Front-side width, back-side width, bead height	Ai et al. (2016)
Laser welding of aluminum alloy (AA5182) with filler wire (AA5356)	Wire feed rate, Laser power, Welding speed	3-5-3-1	GA	Tensile strength	Park and Rhee (2008)
laser–MIG hybrid welding of aluminum alloy (AA8011) with filler material (ER4043)	laser power, welding speeds, wires feed rate	3-11-1	Bayesian regularization	Tensile strength	Chaki (2019)
Laser oscillation welding of aluminum alloy (6061)	laser power, welding speed, oscillation frequency, oscillation amplitude	4-7-1	BP	Weld area	Ai et al. (2023)
Laser transmission welding of clear acrylic and polycarbonate plaques using incremental scanning technique	laser power, scanning speed, pulse frequency, wobble width, wobble frequency	5-11-6-2	non-dominated sorted genetic algorithm	Weld shear strength, weld seam width	Kumar et al. (2024)

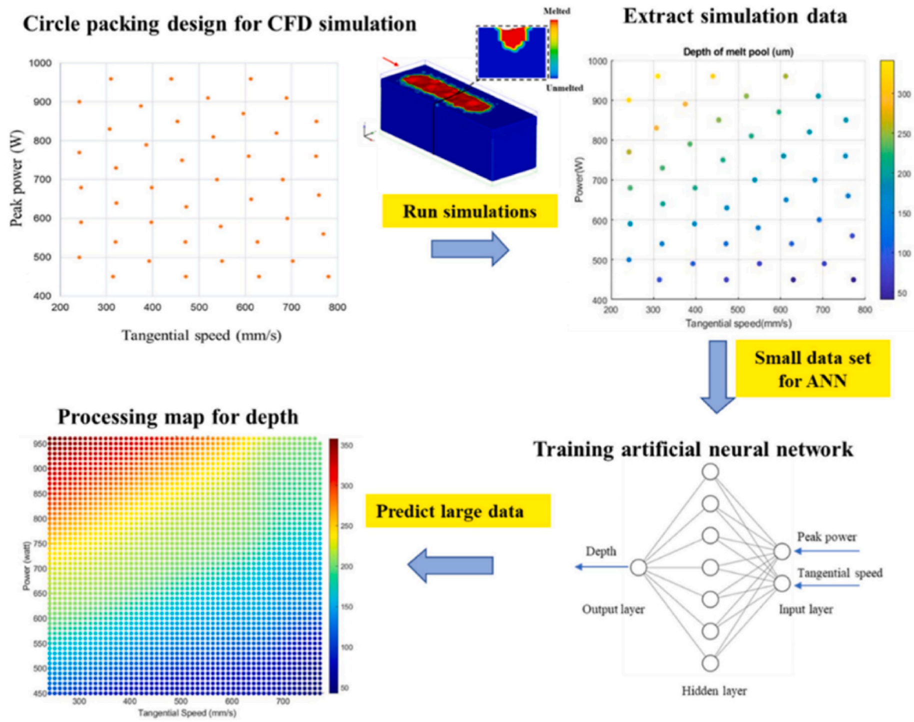


Fig. 5. Surrogate modeling approach for laser welding of dissimilar metal (Thi Tien et al., 2023).

through alterations in welding parameters. This underscores the distinct advantage of utilizing process signals. Among the various dynamic signals, reports on plasma optical signal spectroscopy hold particular prominence. For example, alloying elements, exemplified by aluminum in steel welds, demonstrably exert a deleterious influence on the mechanical properties of the weld (Kang et al., 2012). This phenomenon becomes especially pronounced when workpiece with coatings, representing a significant factor in the degradation of joint quality. Although the ANN, which employs process parameters as input, demonstrates significant superiority in predicting the mechanical performance of joints, it remains insufficient in addressing this challenge. This is due to the fact that the variation of welding parameters is not influenced by the content of alloying elements in the weldment. Based on the hypothesis that spectral signals provide accurate feedback on the fluctuations occurring in the weld area, numerous researches have been conducted.

Research has already established that once the emission lines in the welding plasma spectrum are identified, it becomes feasible to discern the ablation of alloying elements through plasma optical spectroscopic analysis (Mirapeix et al., 2016). The ablation of alloying elements reflects the laser absorption during the laser-matter interaction. (Anabitarte et al., 2012) developed a sensor system founded on a laser-induced breakdown spectroscopy setup. This system is designed to detect and differentiate aluminum residues in the welding region without necessitating sample destruction prior to welding. A spectral algorithm, relying on Support Vector Machines (SVM), functions as a classifier for the automatic identification of regions containing aluminum traces.

Furthermore, the cooling rate, which has a direct impact on mechanical performance, can also be predicted through the energy density monitored by emission spectroscopy. Additionally, due to the influence of cooling rate on microhardness variations, it has become possible to monitor changes in weld hardness through emission spectroscopy. (S. Wang et al., 2020) established a dimensionless nonlinear regression model for iron ion spectral line intensity and dimensionless microhardness using characteristic spectral line calculations. This model proved highly accurate in predicting zone hardness, with maximum and minimum relative prediction errors of 3.09% and 1.6%, respectively. (Zhang et al., 2021) studied the regression prediction of laser welding seam strength of aluminum-lithium alloy by an extreme gradient boosting decision tree (XGBoost) combined with random fores (RF) and principal component analysis (PCA).

Plasma optical signal spectrometry has shown great potential in predicting and optimizing mechanical properties by monitoring chemical composition (Song et al., 2017; Zeng et al., 2023) and thermal effects (Kong et al., 2012; Sibillano et al., 2012). However, the construction and calibration of plasma spectrometry systems are complex, and these systems come with limitations in operational range and substantial costs, which hinder their broader industrial application.

4.1.4. Summary

After conducting a thorough examination of research on mechanical properties prediction in ML-based IPM-LDPW, several key conclusions can be drawn and the comparison of the modeling methods is listed in Table 5:

(1) The mechanical performance of welding is influenced by the temperature gradient and cooling rate, which, in turn, are determined by the absorption of laser energy. Process parameters, laser energy input-related factors, are a strong mechanism

Table 5
Common methods of mechanical performance prediction.

Categories	Features & Advantages	Weaknesses & challenges
Process parameters-based methods	Sensor-independent;	Insufficient robustness to environmental variation;
	Good interpretability;	Low computational resource requirements.
	Small data.	
Process signals-based methods	Reflect the variation in laser absorption; Better model performance;	Demands for relatively massive scale labeled data; Data labels are difficult to produce;
	Features based on sensors have interpretability.	Plasma plume is insufficient for precise monitoring; Robustness depends on the completeness of dataset.

for embedding these physical principles and a strong constraint for modeling based on the hypothesis of an ideally stable welding process. The extent of laser absorption can be indirectly estimated by assessing the chemical composition and thermal effects. Process signals-based methods embed the physical principles by the observational data collected from the welding zone, based on the hypothesis that the laser absorption variation can be well captured by the sensors.

(2) Experiment design methods, including the Taguchi method and RSM, aim at covering a more complete sample space with the least amount of data in this field. These methods achieve a great trial between the completeness of data and the costs.

(3) Machine learning methods, especially ANN, have demonstrated potential in addressing the changes brought by the different laser welding techniques combined with the heterogeneous material properties in the task of mechanical performance prediction and its inverse problems. Multi-task learning introduces the joint optimization of weld geometry and mechanical performance, which effectively improves the stability of the model. Process signals-based methods relying on OE signals have shown promising results in mechanical performance prediction, particularly when alloying elements like aluminum are present in steel welds.

4.2. Weld penetration estimation

Weld penetration refers to projecting the mechanical properties onto a macro-scale geometric feature space. It holds a closer position to the laser energy absorption compared to mechanical performance in the procedures of weld formation. Currently, there are two primary challenges to monitoring weld penetration: the inner molten pool is not visible, and the latent heat exchange during phase transformation is imperceptible. As a problem of interest to all fusion welding monitoring, there has been significant research into using traditional and deep learning methods to solve this problem. Sect 4.2 provides a comprehensive survey of the weld penetration estimation based on ML-based IPM: classic machine learning-based methods, deep learning-based methods, and multimodal fusion-based methods.

4.2.1. Classic machine learning-based methods

Classical machine learning techniques have relied on hand-crafted features as inputs to data-driven models. There have been numerous studies in this field to determine the most effective feature space for accurately predicting weld penetration based on domain knowledge. In this section, we will introduce detailed practices based on these methods.

4.2.1.1. Molten pool and keyhole-based methods. **Spectral-based methods.** These methods aim to classify the weld penetration by analyzing the in-process spectroscopy of OE signals in the reflected laser, VIS, and IR spectral ranges (Angeloni et al., 2024). All the OE signals can reflect the fluid flow on the surface of the molten pool, especially the Marangoni convection, also known as surface tension-driven convection (D. Y. You et al., 2014), which simultaneously influences the width of the molten pool and depth of the keyhole (Wu et al., 2019). However, the OE signals of reflected laser light only can reflect the intense variation in the molten pool (Doong et al., 1991; Shevchik et al., 2019), are not suitable in most situation. Based on this priori knowledge, low-complexity models, such as linear regression (Xiao et al., 2020b), ANN (Yu et al., 2020), k-nearest neighbors (KNN), decision tree (DT), random forest (RF), Naïve–Bayes, support vector machine (SVM), and discriminant analysis (Chianese et al., 2022), can predict the depth and width of weld penetration with OE signals.

Images/videos-based methods. Traditional schemes for camera-based weld penetration prediction firstly extract discriminative and representative features from the molten pool images and then feed the extracted features into machine learning models to predict certain penetration states as multiclass classification problem. Image processing algorithms concentrated on threshold segmentation are designed to extract geometric textural features, including keyhole area (Luo and Shin, 2015a), and molten pool surface area (Lei et al., 2018). Some of the popular techniques are ANN (Lei et al., 2019; Gao et al., 2021; Wu et al., 2024), RF (P. Zhao et al., 2020), SVM (Lu et al., 2020a), KNN (Buongiorno et al., 2022), and DT (Buongiorno et al., 2022).

Specialized sensing signals-based Methods. Besides the previously mentioned signals, based on specialized sensors that can determine the boundary of different phases in the molten pool, weld penetration estimation has been studied. However, compared to EMAT and X-ray, OCT has been well-promoted for industrial applications. However, the presence of the mushy zone and the heat transfer driven by flows inside the molten pool (Wu et al., 2019; Zhang et al., 2018c) pose challenges to the precise measurement of the keyhole profile. In response to this issue, (Stadter et al., 2020) endowed an ANN with prior knowledge from process parameters and accurate prediction of the penetration depth was achieved by simultaneously inputting the 100-D data of OCT.

4.2.1.2. Plasma plume-based methods. **Spectral-based methods.** Spectral monitoring is a well-established technique used for analyzing and observing plasma plumes. The intensity of the spectrum of plasma plumes in the time domain has been used to predict the penetration depth by utilizing techniques such as ANN (Gong and Olsen, 1997; Li et al., 2023), multiple regression (Park and Rhee, 1999), SVM and KNN (Lee et al., 2020). The frequency domain characteristics of the plasma plume are strongly associated with the penetration modes (Zhang et al., 2018b). Additionally, the frequency spectrums of OE signals have demonstrated the ability to sense overlap gaps between weldments by using techniques such as ANN (Beyer et al., 1994). Plasma electron temperature and density are typical indicators for delineating variations in penetration (Chen et al., 2013; Shaikh et al., 2006). Moreover, there is an inherent connection between the electrical properties of plasma and the quality of welds, given that plasma consists of a multitude of free electrons and positive ions (Huang et al., 2019a). (Huang et al., 2019c) established a correlation between plasma electrical signals and penetration by using techniques such as probabilistic neural network and feature extraction methods such as Wavelet Packet Transform (WPT) and Empirical Mode Decomposition (EMD), achieving an average accuracy of 90.16%.

Images-based methods. The morphology of the plasma plume has a significant impact on the depth of penetration during welding. As shown in Fig. 6, the movement of the plasma plume is linked to the oscillation of the keyhole (Wang et al., 2012; Xue et al., 2022). Both the plume inclination spectra and the keyhole area spectra exhibit similar primary frequencies (Tenner et al., 2015). The trajectory of the vapor plume's center in relation to the keyhole position can reflect the instabilities in the welding process (Brock et al., 2013). (Gao et al., 2013) have utilized this regularity to distinguish welding stability using various features, including the coordinate ratio of the plume centroid and the weld point, the polar coordinates of the plume centroid, and several spatter-related features, incorporating the KNN method with the Karhunen-Loeve transform.

Soundwave-based methods. The primary source of AE during laser welding is the ejection of the vapor plume and the force exerted by the plasma plume on the molten pool (Luo et al., 2019). Employing machine learning methods to analyze the time-frequency characteristics of acoustic signals is a prevalent approach in ML-based IPM-LDPW. The Welch–Bartlett method (Huang and Kovacevic, 2009b), Short-Time Fourier Transform (STFT) (Farson et al., 1994), and Wavelet Decomposition (WD) (Luo et al., 2005) have been successfully utilized to extract time-frequency characteristics and distinguish between full penetration and partial penetration welds. To establish the correlation

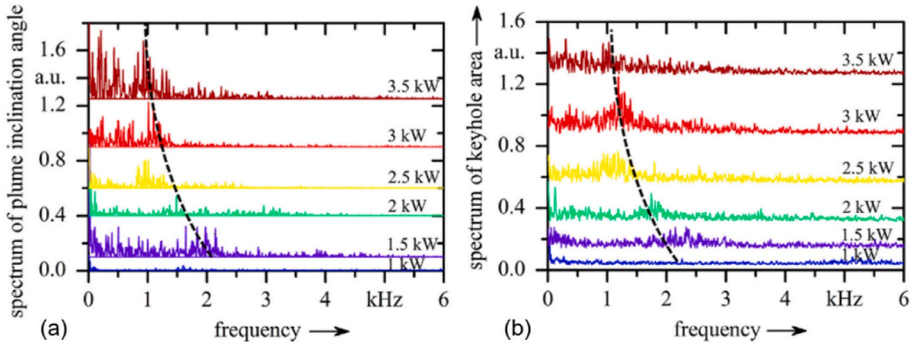


Fig. 6. Spectra of (a) the plume inclination angle and (b) the visible keyhole area (Tenner et al., 2015).

between time-frequency features and weld formation, ANN (Huang and Kovacevic, 2011) and gaussian process regression (Tomcic et al., 2022) are the commonly employed classic machine learning technique.

4.2.2. Deep learning-based methods

This section will give a brief overview of the most common deep networks used in penetration prediction models, including convolutional neural networks, recurrent neural networks, and attention mechanisms.

4.2.2.1. Convolutional models. Convolutional Neural Networks (CNNs) are the building blocks of deep learning architectures designed for visual inference, as they effectively model the spatial structure of images. With the ability to acquire the structural details present in the image and extract more intricate geometric texture features of the molten pool, CNNs have been applied in penetration prediction in form of end to end (Z. Zhang et al., 2020; Xue et al., 2021).

To replace the traditional image process, VGG-16 (Knaak et al., 2019) has tried to differentiate the background, weld seam, weld pool, and keyhole, as shown in Fig. 7 (a). However, in a probabilistic environment, with different equally probable outcomes, pixel-wise losses aim to accommodate uncertainty by blurring the prediction, resulting in the unsmooth contours of the molten pool. To address this issue, U-net (Cai et al., 2022; Fan et al., 2023) has achieved smoother contours of the molten pool and keyhole, as shown in Fig. 7 (b). Based on the textural features extracted by CNNs, classic machine learning methods, such as KNN and RF, and deep learning methods, such as LeNet were sequentially combined for penetration prediction. Compared to traditional image process algorithms, CNNs are much faster and more robust in edge detection.

For time series, by directly inputting of 1-D OE of Cu emission, an accuracy of over 90% in penetration classification has been obtained by CNN, in Al/Cu lap joints used in electric vehicle batteries (Lee et al., 2021). Moreover, converting time series into spectrograms is also a common practice to use CNNs in learning the underlying patterns in AE and/or OE time series data (Drissi-Daoudi et al., 2023; Yanxi Zhang et al., 2019c; Zhang et al., 2022).

Convolutional models also play the role of integrating the observational data of different physical phenomena. In this context, convolutional models aim at learning the underlying features from different multi-fidelity data and aggregating them to predict weld penetration (Shevchik et al., 2020). employed a CNN to extract the relationship between the weld penetration and the wavelet spectrogram of OE and AE signals.

Vanilla CNNs lack the ability to model inter-frame variability in a video sequence. To address this limitation, 3D convolutions have emerged as a promising alternative to recurrent modeling (Wang, 2023). Recently (Yang et al., 2024), designed a two-stream 3D CNN to better capture the spatial and temporal regularities of plasma plume, molten pool, and keyhole in laser welding videos for penetration classification.

4.2.2.2. Recurrent models. Recurrent models were specifically designed to model a spatiotemporal representation of sequential data such as video sequences. Recurrent neural networks (RNNs) have exhibited the potential in sequence modeling, including nature language processing, speech recognition (Miao et al., 2015), and nowcasting forecasting (Shi

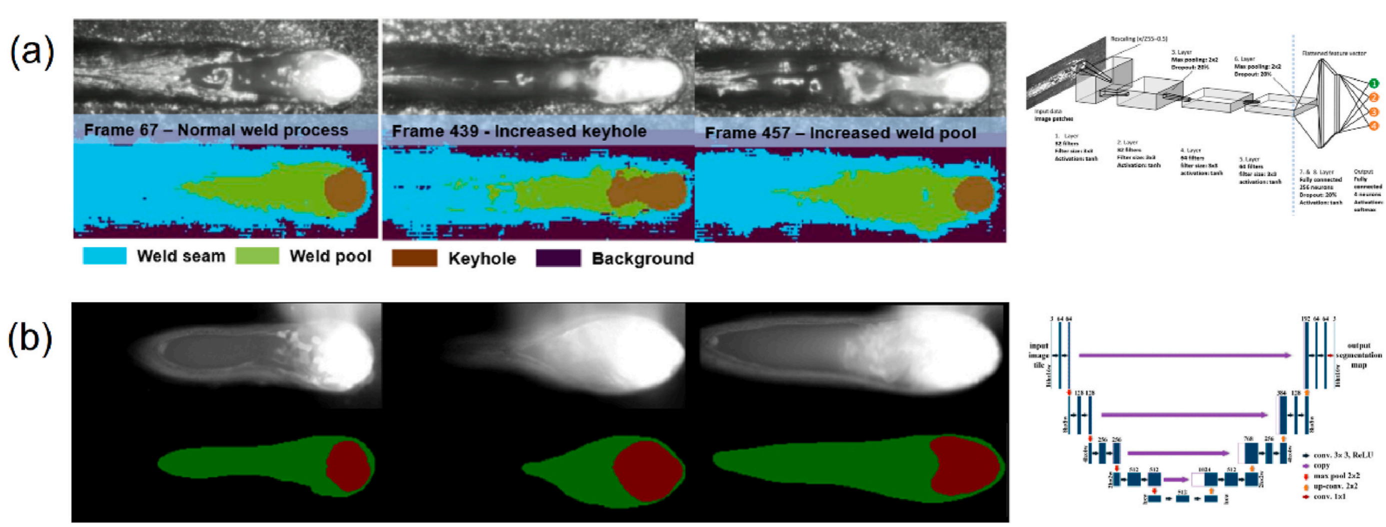


Fig. 7. (a) Structure of the proposed CNN for pixel-wise classification and the results of semantic segmentation (Knaak et al., 2019), (b) Structure of the U-net and CNN combined model for penetration status classification (Cai et al., 2022).

et al., 2015). However, vanilla RNNs suffer from vanishing and exploding gradients, limiting their effectiveness in capturing long-term dependencies (Zaremba et al., 2014). Additionally, RNNs are time-consuming due to their serial structure and large number of parameters (Tu et al., 2018). Many researches have conducted to address these issues, such as Long-Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997), PredRNN(Y. Wang et al., 2023), and PredRNN++ (Wang et al., 2018),etc.

Recurrent models have shown the potential in aggregating the spatial, temporal, and thermal information by the memory mechanism. Assuming that the variable characteristics of the plasma plume, and the molten pool and keyhole can be learned in the time domain by the recurrent architectures, gated recurrent unit (GRU) (Mucllari et al., 2023), CNN-GRU (Knaak et al., 2021), CNN-LSTM (Yu et al., 2022; Luo et al., 2023), and attention-LSTM (Ye et al., 2022) have been carried out in penetration prediction.

4.2.2.3. Attention models. Attention mechanism is an essential component of large models (Vaswani et al., 2017). There are three typical types of attention mechanisms: self-attention mechanisms, spatial attention mechanisms, and temporal attention mechanisms. These mechanisms enable the model to allocate varying weights to different locations within the input sequence. As a result, the model can focus on the most relevant parts when processing each sequence element.

Enlightened by this, (Liu et al., 2022) designed a label semantic attention mechanism to guide the CNNs in penetration prediction, which obtained a faster convergence and higher accuracy than the traditional attention mechanism. Additionally, by taking cues from Transformer-based architecture, it becomes possible to employ more flexible Self-attention layers that can capture long-range dependencies within long time series data. (Deng et al., 2022) have proposed a CNN that integrates window-based transformer blocks to classify weld quality. This approach has been shown to outperform existing CNN and Transformer methods, while also being more parallelizable and requiring considerably less training time.

Besides that, the informer is an architecture based on the Transformer model that has exhibited excellent performance in tasks previously handled by RNNs. (Fan et al., 2023) used an informer model to predict weld penetration by combining current and historical welding information, which resulted in better performance compared to GRU and LSTM models. Additionally, (Cao et al., 2023) showed that cross-attention mechanisms can effectively fuse information from AE and OE signals processed by convolutional layers in different streams for predicting weld penetration.

4.2.2.4. Other methods. **Interpretability techniques.** To comprehend the insights derived by deep learning methods in the laser welding process, advanced interpretability techniques have been employed in ML-based IPM-LDPW research. For instance, Gradient-weighted class activation mapping (Grad-CAM) can visually display the features learned by the CNN, which helps to understand the working principle and decision-making process of the neural network (Das and Ortega, 2022). Božič et al. (2020) employed the Grad-CAM to comprehend the decision-making process of the CNN model. The findings revealed that the CNN primarily concentrates on the interaction zone and can adapt its performance based on any changes in its size, as illustrated in Fig. 8.

Generative models. Learning the probability density of observable samples, as well as new samples generated by generative models, is the fundamental principle of Larger Language models (LLM). In this field, generative models have been used for data augmentation (Fan et al., 2021) and microstructure prediction (Oh and Ki, 2020). conducted groundbreaking research that utilized conditional generative adversarial networks (cGAN) to predict weld penetration. The research incorporated several components such as weld geometry, the heat-affected area, and microstructure. The research featured two generators, each

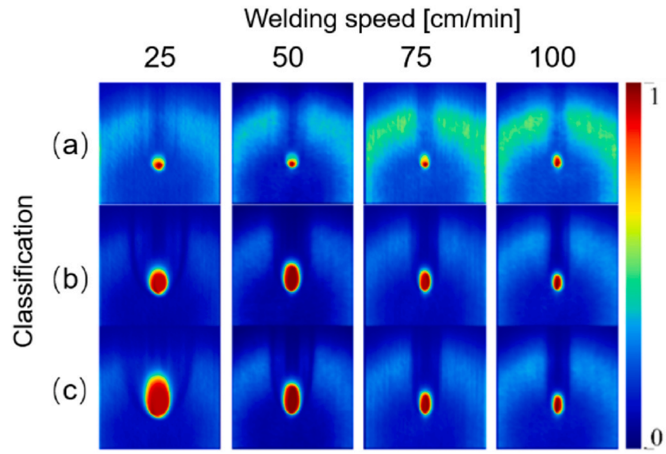


Fig. 8. Result of the Grad-CAM of the output of the first convolutional layer, where red means that the pixel has an impact on decision making and blue means that the pixel has no impact (Božič et al., 2020). Rows present categories for (a) too little, (b) sufficient, and (c) exceeding energy input.

consisting of two encoder-decoder models based on Convolutional Neural Networks, as shown in Fig. 9. The first generator takes two-channel maps of laser intensity and interaction time as inputs, while the second generator has additional input in the form of weld contour segmentation for the discriminator. The corresponding textual information is directly provided to the connection layer as control conditions for the first generator, contributing to the control of weld bead profile generation. Finally, the research team accomplished the prediction of welds by incorporating the process parameters in text-to-image format. While the grain details inside the weld bead were not fully realized, the application of deep learning in this field has shown great potential for further research.

4.2.3. Multimodal fusion in weld penetration estimation

The use of signals collected from multiple sensors in Machine Learning-based In-Process Monitoring (ML-based IPM-LDPW) is an example of multimodal machine learning. This approach yields better classification performance as compared to using a single-modal feature in isolation. (Atrey et al., 2010; Baltrušaitis et al., 2019). However, it brings forth five primary challenges: representation, translation/mapping, alignment, fusion, and co-learning (Baltrušaitis et al., 2019).

In the conventional implementation of multimodal machine learning in ML-based IPM-LDPW, various signals are collected during the same laser welding process. These signals include optical, acoustic, electrical, and thermal signals. The most commonly used methods for representation are PCA and wrapper methods, which enhance complementarity and remove redundancy between different signals. Translation/mapping is also a key aspect of multimodal machine learning for monitoring and diagnosing LDPW. This involves tasks such as converting images of the molten pool and keyhole into a time series of geometric features and transforming time-domain signals of AE into frequency-domain signals.

Multiple sensors are simultaneously activated to acquire physical information during laser welding. (Yanxi Zhang et al., 2019c; Liu et al., 2019; Dold et al., 2023). Therefore, multimodal alignment in ML-based IPM-LDPW is rooted in time dimension alignment. Early fusion is the most commonly used fusion method. Generally, various types of sensors are used to gather process signals. Features are extracted from the collected signals using physical principles and statistical methods, and then merged using machine learning models (Dold et al., 2023; D. You et al., 2014b).

Multimodal machine learning is commonly used in the monitoring and diagnosis of LDPW. Due to the time-consuming nature of data labeling, transfer learning is an effective co-learning method for

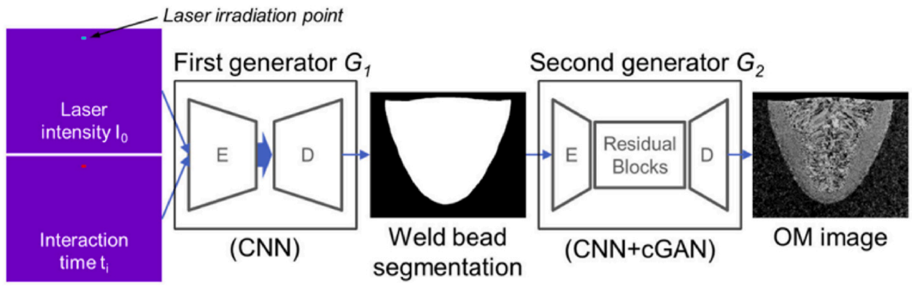


Fig. 9. cGAN-based framework of weld penetration prediction (Oh and Ki, 2020).

multimodal machine learning in LDPW monitoring (Yang et al., 2020).

4.2.4. Summary

Table 6 provides an overview of the notable implementations of ML-based IPM-LDPW specifically focusing on weld penetration. Table 7 provides the critically analyze for the classic machine learning-based methods and the recent deep learning-based methods. The following key observations can be made from this section:

(1) Both classic machine learning and recent deep learning techniques have been successfully employed for penetration estimation, yielding commendable results. However, in general, deep learning approaches have exhibited superior accuracy.
(2) Considering the prevalence of signal applications, the hierarchy of utilization of different process signals can be ranked as follows: Image > Spectral > Process parameters > Soundwave. Notably, acoustic sensors display substantial potential for application.
(3) The integration of multimodal machine learning emerges as a promising future avenue for ML-based IPM-LDPW. For instance, (D. You et al., 2014b; Dold et al., 2023) effectively employed multimodal signals, combining vision and spectrum data, to achieve precise penetration estimation.
(4) Under controlled experimental conditions, the accuracy of penetration estimation currently attains an impressive 99.96% (Liu et al., 2022). However, it is imperative to acknowledge that the complexity of real-world conditions may restrict the applicability of the trained model. Consequently, acquiring data that encompasses a wide array of operating conditions is imperative. Additionally, the potential shift from macroscopic to microscopic level prediction tasks, as suggested by the results of weld microstructure generation using cGAN (Oh and Ki, 2020), presents a viable research direction.

Table 7
Common methods of weld penetration estimation.

Categories	Features & Advantages	Weaknesses & challenges
Classic machine learning-based methods	Mature theories and models can be used as reference; Physical mechanism-based feature engineering can provide the interpretability; Good robust.	Relies on manual experience or domain knowledge; Feature extractions based on traditional image processing and spectrum calculation are time-consuming; Relatively poor model performance; Prediction accuracy depends on the size and scope of the data used for training.
Deep learning-based methods	Modeling with raw data, no need for hand-crafted features; Can extract features that are unable to feature engineering based on prior knowledge; Better model performance.	Demands for massive scale labeled data; Lack interpretability; High computational resource requirements; Easy to overfit.

Table 6
Recently typical methods of ML-based IPM-LDPW in the task of penetration estimation.

Data type	Sensors	Inputs	Outputs	Machine learning	Performance	Ref.
Spectral	Photodiodes	OE (380–760 nm, 1030 nm)	Weld statuses	SAE	Acc: 92.2 %	(Yanxi Zhang et al., 2019a)
Image	High-speed camera (Photrom Fastcam SA4)	Keyhole and full penetration hole	Weld statuses	SVM RF	Acc: 96.5% Acc: 98.85%	Lu et al. (2020b) 2020/(P. Zhao et al., 2020)
Image	High-speed camera (Phantom V611)	Molten pool and keyhole geometry	Weld statuses	CNN, U-net	Acc: 98.68%	Cai et al. (2022)
Image	High-speed camera (CP70-2M)	Molten pool and keyhole area	Weld statuses	Attention -CNN	Acc: 99.96%	Liu et al. (2022)
Image sequences	High-speed visual camera system (VW-9000E)	Arc, molten pool and keyhole	Weld statuses	Attention-LSTM	MSE: 0.0848, R^2: 0.5900, and CORR: 0.7694	Ye et al. (2022)
text and image	High-speed CMOS camera (Optronics)	Molten pool morphology	Weld geometry	ANN, PCA	MAPE: less than 0.01, MSE: less than 0.001	Lei et al. (2019)
Spectral and image	High-speed camera and photodiode	OE (400 nm–900 nm), and keyhole	Weld formation	WPD-CNN	Average error: 6.12%	(Yanxi Zhang et al., 2019d)
Audio	ABAE sensor (BSWA TECH)	AE	Weld statuses	CNN-LSTM	Acc: 99.8%	Luo et al. (2023)
Image sequences	High-speed camera (768x768 pixels, 2000 FPS)	Molten pool width	Weld width	Informer, U-net	MSE:0.0055/0.1/0.0152	Fan et al. (2023)
Time series	Photodiode and microphone	AE and OE	Weld statuses	Cross-attention fusion NN	Acc + std:99.73±0.37%	Cao et al. (2023)
Image sequences	Two cameras (Imaging Source DMK 33UX273)	Molten pool and plasma plume	Weld statuses	3DCNN	Acc:98.9%	Yang et al. (2024)

'RE' represents relative error, 'Acc' represents accuracy, 'MSE' represents mean squared error, MAPE' represents mean absolute percent error, 'R^2' represents determination coefficient.

4.3. Weld defects detection

Weld defects can significantly impact the mechanical, creep, and fatigue properties of welded joints. There are two major categories of welding defects: appearance defects and internal defects. Appearance defects are caused by the flow of molten metal, which can result in spatter, indicating flow fluctuations in the molten pool (Wu et al., 2022). This results from the molten metal's momentum, either exceeding or falling short of surface tension, leading to upward or downward movement (Li et al., 2020; Zhang et al., 2013). However, predicting internal defects like porosity, cracks, and lack of fusion formation through modeling the welding process remains a challenge, as no signals are collected from the interior of the molten pool where these defects occur (Cai et al., 2020; Wu et al., 2022; D. Y. You et al., 2014).

Indeed, predicting complex welding defects is akin to tasks like image captioning or sentiment analysis, but more formidable, as it involves making inferences based solely on the limited information within the fusion area and an incomplete prior knowledge during LDPW. In this section, we concentrate on the current ML-based IPM-LDPW methods specifically designed to identify pores, cracks, and spatters.

4.3.1. Porosity prediction

These methods establish correlations between information related to pores, including the plasma spectrum and the dynamic behaviors of the molten pool and keyhole, with the presence of pores inside the weld. Based on the type of pore formation, it can be divided into two categories: metallurgical pores monitoring and keyhole-induced pores monitoring.

4.3.1.1. Background. Pores refers to the presence of voids or cavities in the weld metal that result from the entrapment of gas during solidification. These pores or voids have a detrimental effect on the mechanical properties of the weld by reducing the effective cross-sectional area and causing stress concentration. This reduction is especially significant in the tensile strength (Harooni et al., 2014b) and fatigue life (Eibl et al., 2003).

Much like how there are no identical leaves in the world, the distribution of pores in welds under the same process conditions cannot be exactly replicated. The distribution of pores in a weld is primarily influenced by the dynamics of the molten pool. Some elements found within pores may not even exist in the base metal. The fluid flow of the molten pool during welding can make these elements measurable. Moreover, accurately tracking the precise location of pores in the weld may be challenging, but it is possible to monitor and characterize changes in the flow of the molten pool during stable welding. Hence, combining prior knowledge of the spectral line intensity of specific elements or the molten pool flow, along with its driving factor—the heat input, represents a feasible approach for conducting online porosity monitoring.

Pores formed during laser welding can be classified primarily into two types based on the source of the entrapped gas: metallurgical pores and keyhole-induced porosity. Metallurgical pores origin from hydrogen pores (Mikucki and Shearouse, 1993), pre-existing pores, surface coatings on the sheet (Pastor et al., 2000), and pores formed by alloy elements with a low vaporization point (Harooni et al., 2014b). Keyhole-induced porosity, on the other hand, is formed when gas trapped by the collapse of an unstable keyhole is unable to escape from the weld area (Matsunawa et al., 1998). Porosity is a notable concern during laser welding of Mg or Al alloys, as shown in Fig. 10 (Huang et al., 2023).

Currently, inner pores detection is performed post-welding, using non-destructive techniques (NDT) such as X-ray tomography (Boley et al., 2013), ultrasonic testing, and magnetic induced analysis (Kah et al., 2014). Indeed, these post-welding non-destructive testing methods are notably time-consuming and labor-intensive, rendering them unsuitable for large-scale industrial production. Consequently, the imperative need arises for the development of in-process monitoring methods relying on porosity-related signals to furnish real-time feedback, especially in the context of mass industrial production.

4.3.1.2. Metallurgical pores. Methods for monitoring metallurgical porosity, encompassing hydrogen and zinc pores, have been devised through the tracking of spectral emissions during welding processes. Determination of electron temperature and electron density through spectroscopic analysis of plasma spectral intensity stands as one of the longstanding applications in plasma physics (Torres et al., 2003). These metrics serve as critical data in the prediction of welding defects (Griem, 2005). Based on the assumption that the laser-induced plasma with a homogeneous distribution should be optically thin and in local thermal equilibrium for the laser welding process, (Harooni et al., 2014a) extracted the electron temperature by processing the spectra emitted from a plasma plume using the Boltzmann plot method, the results showed good correlation between the pore formation and the spectral

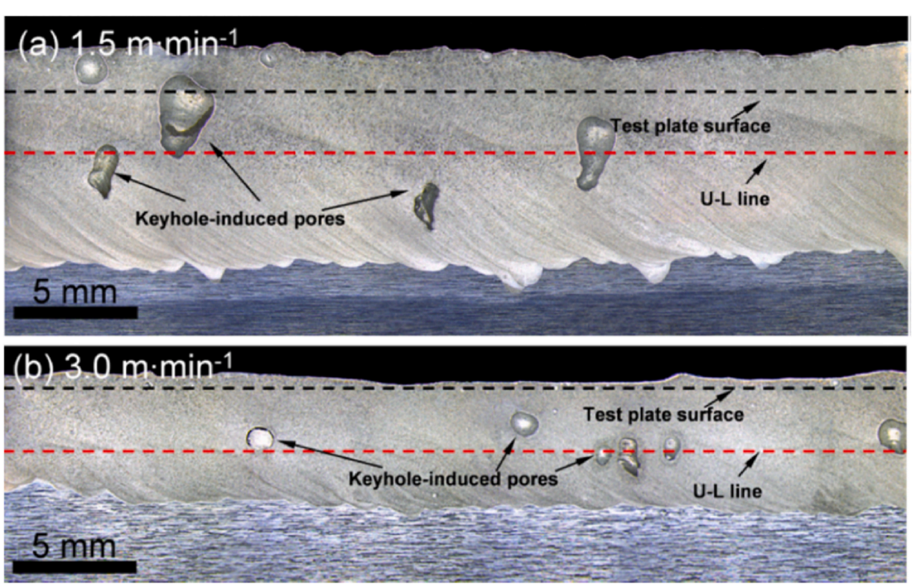


Fig. 10. The views of the weld longitudinal sections show the distribution of pores at the different welding speeds:(a) 1.5 m min^{-1}, (b) 3.0 m min^{-1} (Huang et al., 2023).

signals. Spectral data often has a lot of dimensions, and the spectral bands are highly correlated with each other. This leads to a lot of redundant information, especially since the neighboring bands tend to be strongly correlated. As a result, the data is often very sparse and irregular when plotted in high-dimensional space. PCA (Colombo et al., 2013) has been successful applied in reducing dimensionality of emission spectrum for tracking the overlap gap between plates, and the gap is correlated to the pore formation. Other dimensionality reduction algorithms, such as Singular Value Decomposition, Stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002), t-SNE(Kobak and Berens, 2019), and universal Pattern Decomposition Method (Zhang et al., 2007), can also be used for spectral data as well.

Research focused on monitoring the spectral line intensity of specific elements to predict metallurgical pores during laser welding is relatively scarce. Therefore, it would be advantageous to explore and draw insights from studies in similar domains. For instance, in gas tungsten arc welding, (Yiming Huang et al., 2020a; 2020b) employed the local linear embedding and wavelet packet transform (LLE-WPT) to extract features from the arc spectral signal. With the extreme learning method, accurate and swift pores detection was achieved. Similarly, in laser additive manufacturing, (Paulson et al., 2020) explored bunches of machine-learning approaches, including logistic regression, RF classification, gradient boosting classification, and gaussian process classification, to establish correlations between thermal history and subsurface pores under various printing conditions in laser powder bed fusion. The labeling of pores in samples was accomplished through synchrotron X-ray imaging.

4.3.1.3. Keyhole-induced pores. Methods for detecting the keyhole-induced pores in ML-based IPM-LDPW are based on the physical mechanism of keyhole-induced pores. The primary factor responsible for pores formation is the opening keyhole, which depends on the properties of the molten metal, such as surface tension and vapor pressure (Huang et al., 2022; Matsunawa et al., 1998). As shown in Fig. 11, the critical instability at the moving keyhole tip generates pores during laser melting (C. Zhao et al., 2020). Therefore, sensible physical phenomena, including keyhole geometry (Luo and Shin, 2015b), surface texture of the molten pool and keyhole (B. Zhang et al., 2020), and inner geometry of the keyhole (Ma et al., 2022a), have been investigated. Porosity monitoring can be achieved by supervised ANN with the PCA compressed keyhole opening morphology characteristics and the real-time measurement of the keyhole depth as inputs (Ma et al., 2022a).

Plasma plume can provide insight into the dynamics of the keyhole geometry, as depicted in Fig. 12 (Brock et al., 2014). The high transition probability FeI emission lines can correlate to the porosity and bead separation by ANN and SVM (Chen et al., 2019). Furthermore, AE signals associated with the fluctuations of the plasma plume and the keyhole (Tempelman et al., 2022). Acoustic waves, generated by the critical instability of the keyhole in the molten pool, play a crucial role in driving pores near the keyhole tip away from the large thermal gradient field surrounding the keyhole (C. Zhao et al., 2020). By designing features, including peak amplitude, rise time, duration, energy, and the number of counts, the prediction of pores and cracks can be achieved by logistic regression and ANN (Gaja and Liou, 2018).

Leveraging the complementary advantages of the multi-fidelity data, OE and AE have been explored on porosity monitoring. By using Laplacian graph support vector machines (LapSVM) combined with the M-band wavelets decomposition, (Shevchik et al., 2019) has successfully established the correlation between porosity and signals of the OE from the molten pool surface and the AE signals of the volumetric fluctuations of the material phases.

However, as the formation of pores inside the weld is complex, speculating on the results of intercepted pores within the melt pool based solely on external information of the molten pool is insufficient. To address this challenge, deep learning methods have been explored. Some of the methods include convolutional models-based approaches like AlexNet with weld-pool images (B. Zhang et al., 2020), CNN with wavelet spectrograms of AE and OE (Shevchik et al., 2020), and CNN with spectrum graphs of OCT data and illuminant enhanced images (Ma et al., 2022b). To enhance the embedding of physical principle, a multi-fidelity deep learning framework has been proposed to enhance the performance of models by integrating numerically simulated data with sensor-captured data for online porosity monitoring (Ma et al., 2022d). This framework includes a sparse auto-encoder (SAE) and two deep belief networks, as shown in Fig. 13. The results indicate a significant improvement in porosity prediction by using features that are automatically extracted by the SAE from the simulated maximum temperature on the bottom wall of the keyhole and the OCT-measured keyhole depth.

4.3.2. Cracks prediction

Predicting cracks is a much more difficult task than predicting pores.

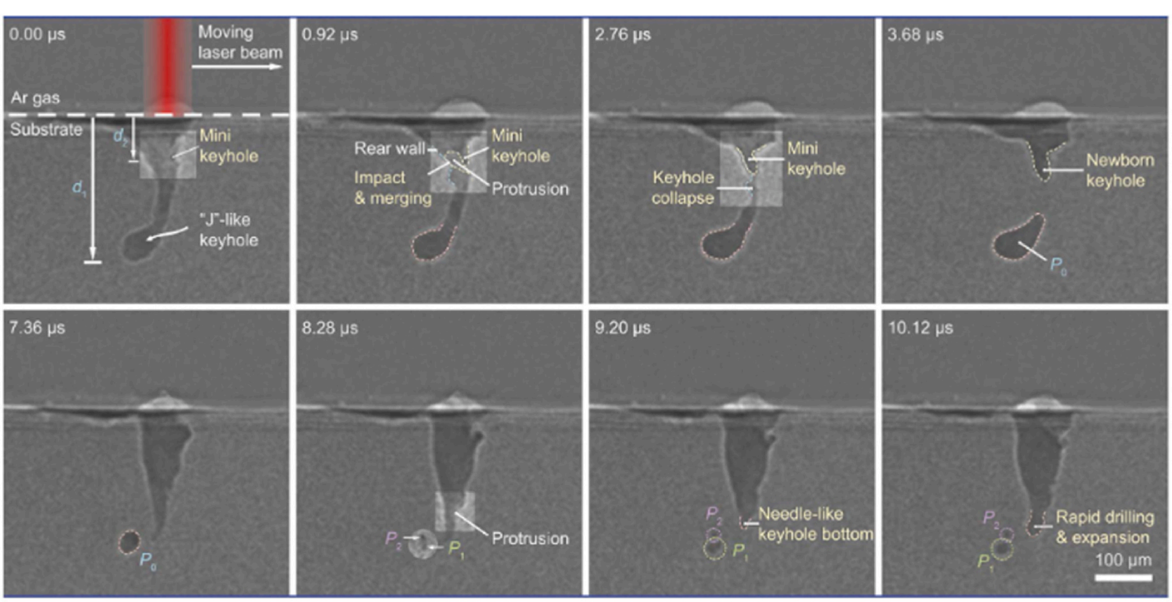


Fig. 11. Megahertz x-ray images of a keyhole pore-formation process (C. Zhao et al., 2020).

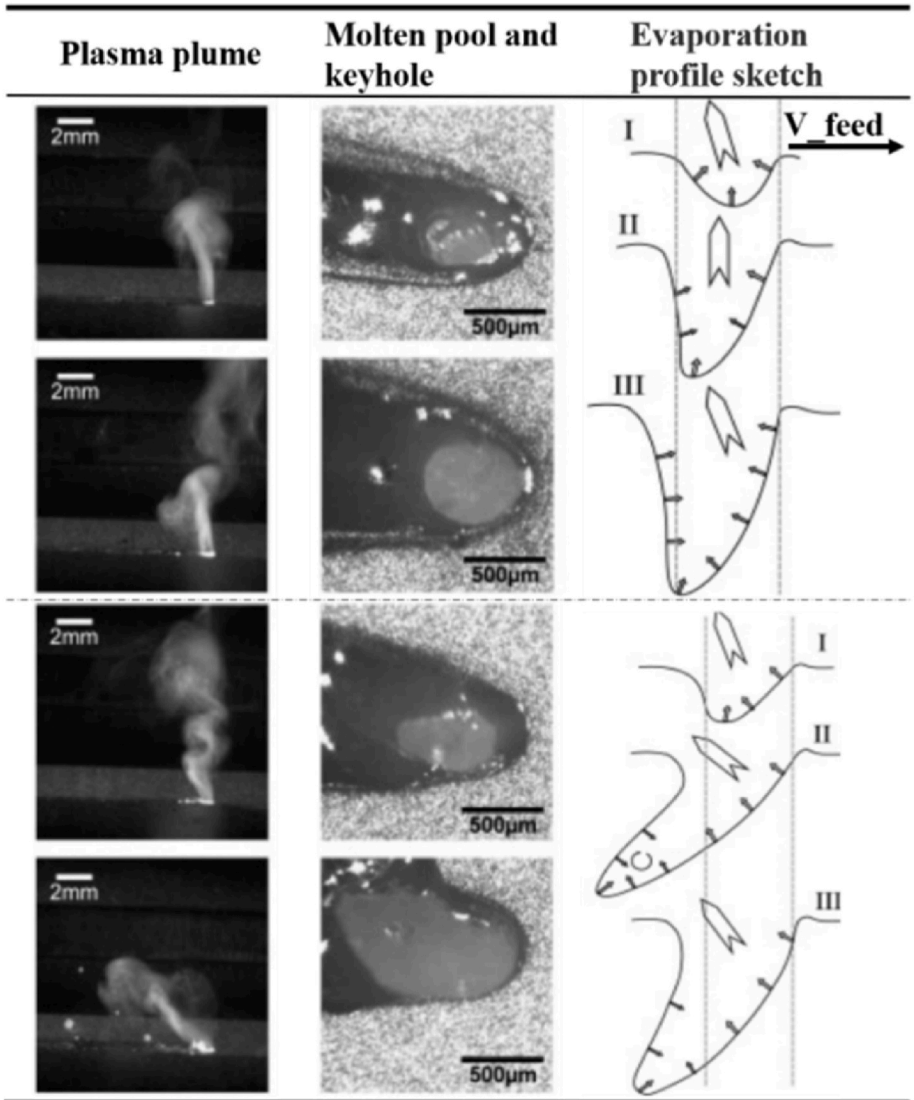


Fig. 12. Schematic diagram of the dependence of the plume inclination on the keyhole geometry (Brock et al., 2014).

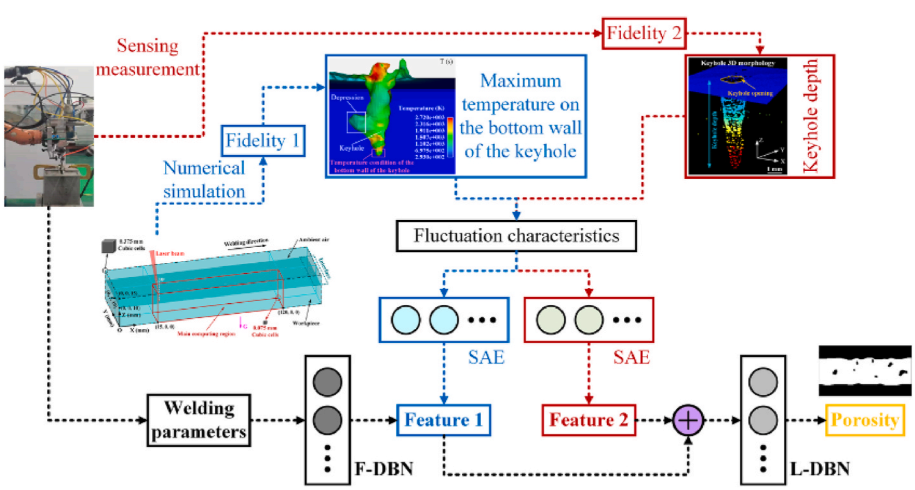


Fig. 13. Multi-fidelity deep learning framework for porosity prediction (Ma et al., 2022d).

This is because cracks are mainly caused by the forces that occur during the formation of microstructures, such as the shrinkage and separation of grains caused by tensile deformation (Kou, 2015). These forces are not directly observable, which makes it a major challenge to accurately predict the occurrence of cracks.

4.3.2.1. Background. Rapid melting and subsequent swift solidification can give rise to cracks in the welded joint, a phenomenon known to substantially compromise both mechanical properties and serviceability (Z. Liu et al., 2023; Mondal et al., 2022). Hot cracking, a common occurrence in welds, is typically attributed to the contraction of weld metal during cooling (Sheikhi et al., 2015). This issue is particularly problematic in the laser welding of magnesium alloys (Stolbov et al., 1991), aluminum alloys (Ghaini et al., 2009), or dissimilar alloys (Z. Wang et al., 2020). Both solidification cracking and liquation cracking are types of hot cracking that can occur in the weld metal and base metal, respectively (Ghaini et al., 2009). Various factors contribute to the formation of cracks. Temperature gradients and solidification rates are recognized to exert a substantial impact on solidification morphology and the scale of microstructure. Moreover, accumulated stress during solidification can significantly expedite the propensity for crack formation (Williams and Singer, 1968).

4.3.2.2. In-process monitoring of crack. Generally, cracks exist inside the weld metal and are almost impossible for sensing the formation process from surface OE signals of molten pool and keyhole. However, the temperature distribution and thermal history of the molten pool surface furnish an abundance of physical information concerning the temperature gradients and solidification rates, making the cracks prediction possible. To tackle the challenge of implementing IPM for cracks, drawing from this understanding, von Witzendorff et al. (2015) conducted simultaneous observations of the temperature profile and size of the molten pool using both an infrared and a visible camera during laser pulse welding of 6082 aluminum alloy. Their results demonstrate that hot cracking typically arises in the final stages of weld solidification, and that temperature evolution curves and molten pool development can serve as effective predictors of crack formation.

Furthermore, AE has been demonstrated to capture the thermal stress and cracking that occur during the solidification of the weld pool (Zeng et al., 2001). Furthermore, the AE signal can serve as a means to monitor the rapid phase change between liquid and solid (Sun et al., 1999). (Jia et al., 2021) monitored the pulse laser welding process and effectively captured the propagation of cracks via high-speed photography and acoustic emission, as depicted in Fig. 14. (W. Liu et al., 2023a) discovered that the envelope area of the energy curve in the tail band remained stable in most cases, but exhibited significant changes when large cracks occurred. The energy curve's characteristics in the tail band also indicated that crack generation took place after laser ablation. AE signals comprise multiple information sources, making the direct identification of pertinent information regarding crack formation difficult. Signal processing methods, such as WPT (Shevchik et al., 2020), Hilbert–Huang transform (HHT) (Lin and Chu, 2011), Local Mean Decomposition (LMD) (Yanqiang and Jie, 2017), cosine similarity (W. Liu et al., 2023b), and others have been applied to extract characteristic signals.

Machine learning techniques have been utilized to understand the components associated with crack formation. (Lee et al., 2014) utilized an ANN to differentiate the frequency spectrum of AE signals, revealing that AE events may indicate potential crack initiation mechanisms. In addition to the features of AE, process parameters and material properties play crucial roles in predicting crack formation. (Mondal et al., 2022) applied physics-informed machine learning (PIML) to achieve crack-free metal printing. A mechanistic model was utilized to compute four physical variables related to cracking: solidification stress, ratio of vulnerable and relaxation times, solidification morphology, and cooling rate, as demonstrated in Fig. 15. These variables, along with experimental data, were used to generate a cracking susceptibility index that predicts crack formation before printing. Machine learning techniques such as SVM, DT, and Logistic Regression can accurately predict crack formation. Additionally, "easy-to-use" cracking susceptibility maps were derived from the prediction results.

4.3.3. Spatters recognition

Detecting spatters during the post-process monitoring stage can be a difficult task for anomaly detection. The reason being that spatter's color is similar to the workpiece, and the texture is similar to the weld. However, this challenge can be tackled by using IPM (Image Processing Methods). This is because the process of spatter generation is quite evident. Therefore, Machine Learning-based IPM for spatters is considered an object detection task. This is because spatters can be directly observed during the welding process. In this section, we will discuss the principles of spatters, both conventional and ML-based methods, and the application of spatters.

4.3.3.1. Background. Spattering is a critical problem that occurs in various fusion welding techniques. It can have negative effects on laser energy transmission in laser welding, leading to the expulsion of molten droplets from the melted pool. This can cause welding defects like underfill, undercuts, craters, blowholes, or blowouts, which can significantly impair the performance of the welded joint (Kaplan and Powell, 2011). Researchers have conducted extensive research to understand the mechanisms behind spatter formation. Spatter formation is usually initiated by the recoil pressure that disrupts the force equilibrium of the keyhole wall, surface tension pressure, and fluid flow (Zhang et al., 2018d). This creates a robust steam plume shear flow that vigorously propels the melt along the keyhole walls (Nakamura et al., 2015; Zhang et al., 2013). When the downward or upward momentum of the melt surpasses the surface tension, it leads to the formation of front or back weld spatters (Li et al., 2020; Zhang et al., 2013), as depicted in 17 (a) (D. You et al., 2014b). These events occur dynamically, including the severe injection of plasma and metal vapor, as well as violent oscillation of small holes in the molten pool, manifesting as clear outliers in the data distribution.

4.3.3.2. In-process monitoring if spatters. The presence of spatters is an unusual occurrence that can be identified through various methods. AE

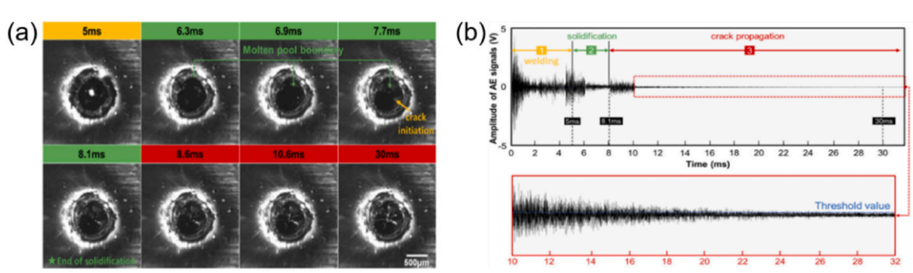


Fig. 14. Monitoring of cracks during single pulse welding process. (a) High speed photography; (b) Amplitude of AE signals (Jia et al., 2021).

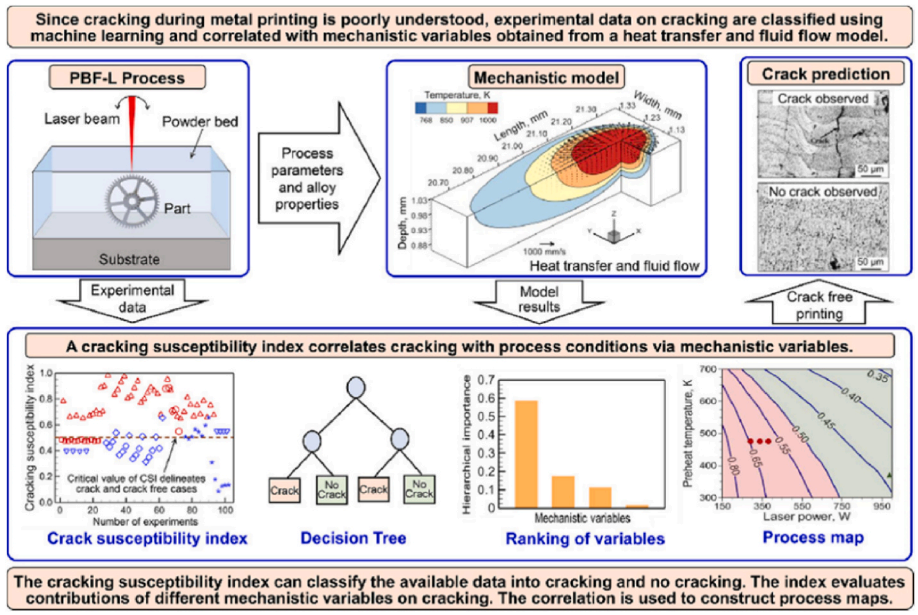


Fig. 15. Physical-informed machine learning for crack free metal printing (Mondal et al., 2022).

and OE are two techniques commonly used for recognizing spatters. In a study conducted by (Ye Huang et al., 2020), a high-speed camera was used in combination with motion tracking to analyze the frequency density distribution of spatters. The outcome was depicted in Fig. 16 (b), and it was found that the spectrum of these signals can better identify the occurrence of spatters. (Yang et al., 2009) utilized STFT to extract the AE of spatters, and the results indicated that a sharp spike with high amplitude indicates the formation of spatters. This was demonstrated in Fig. 16 (c). OE in the infrared range has also proven to be effective in spatter monitoring (Park et al., 2001). To investigate the feasibility of using these techniques in industrial applications, (You et al., 2015) explored the substantial influence of the low-frequency component of the photodiode signal in the infrared/VIS range on the detection of weld defects, particularly spatters. The results showed that a feed-forward neural network (FNN) utilizing both the photodiode and spectrometer signals as inputs can be a dependable method for monitoring spatters. In end-to-end learning, (Schmidt et al., 2020) employed a CNN to denoise and extract weld quality information during laser welding. The results showed the potential possibility of spatters detection by using AE signals.

4.3.3.3. Application of spatters-based features. Although spatter may seem unattractive and have the potential to cause welding defects like porosity, it can also serve as a useful indicator of welding stability. By treating spatter levels as a process feature, we can gain valuable insights into the performance of the welding process. (Gao et al., 2014) proposed a method that uses an ANN to predict weld bead width. This method takes into account the number and area of spatters, as well as the area, height, tilt angle, and centroid of the plume as inputs to predict the weld bead width. (Chen et al., 2018) used features extracted from the vapor plume and spatters to classify weld penetration by utilizing SVM. The experimental results showed that the SVM classifier, using seven selected features, achieved a desirable accuracy of 95.93% through a 10-fold cross-validation. (Yanxi Zhang et al., 2019b) used the spatter number as an input to a Deep Belief Network (DBN) model to detect welding defects such as blowouts, humps, and undercutting. Additionally, they designed a DBN structure based on a stacked sparse autoencoder, as illustrated in Fig. 17, to better explore the features, including the spatter number presented in their previous work (Y. Zhang et al., 2020).

4.3.4. Summary

Table 8 presents the typical methods utilized in IPM-LDPW for weld defect detection. Comparison between the methods is listed in Table 9. Several key observations can be made:

(1) ANN are commonly employed to establish correlations between process signals and weld defects. They have demonstrated excellent performance in detecting various types of defects.

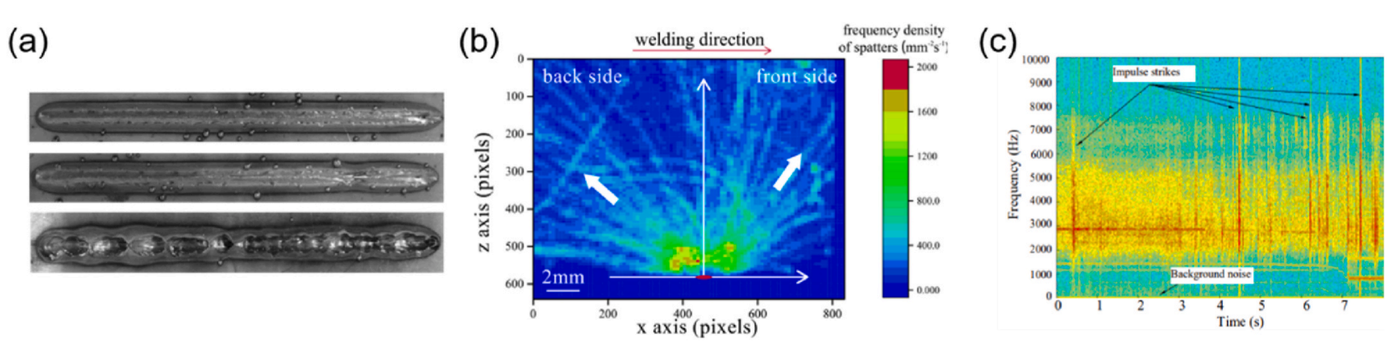


Fig. 16. Spatters defects on the workpiece and its typical monitoring results. (a) Spatters motion during laser welding process (D. You et al., 2014b), (b) frequency density distribution of spatters (Ye Huang et al., 2020), STFT processed AE of spatters (Yang et al., 2009).

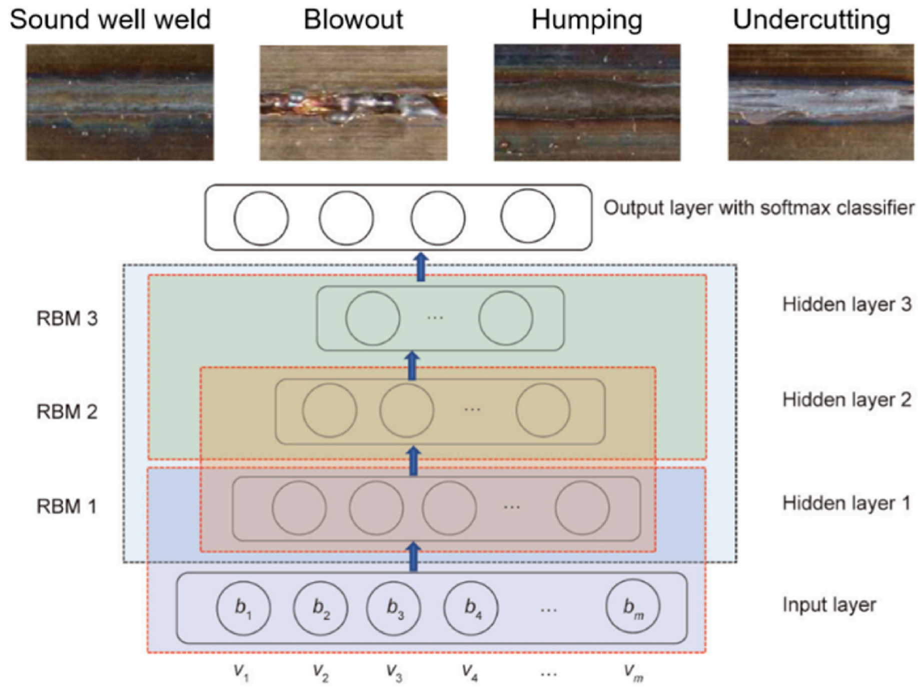


Fig. 17. The designed structure of the DBN (Yanxi Zhang et al., 2019b).

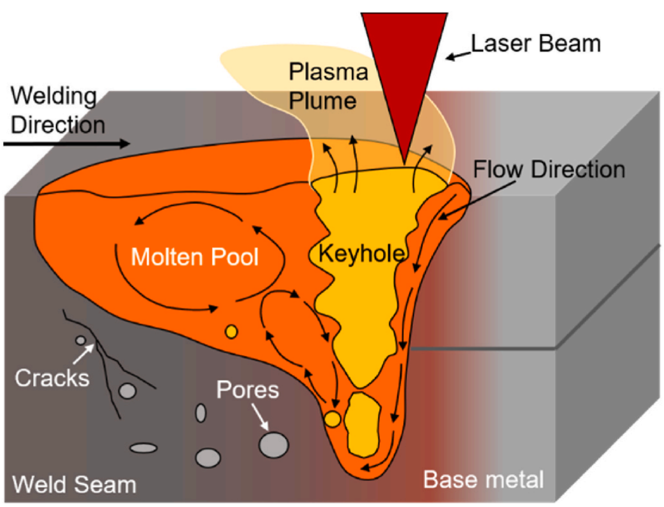


Fig. 18. Schematic representations of the formation of pores and cracks.

(2) OE and AE are commonly employed in detecting weld defects due to their direct correlation with defect formation. However, it's worth noting that AE possesses even greater potential in identifying imperceptible and internal weld defects. This highlights the critical importance of thoroughly investigating the relationship between weld defects and AE signals.
(3) PIML harnesses the underlying mechanisms of weld defects to overcome the limitations of process signals, showing promise in detecting otherwise imperceptible weld defects with good interpretability. Nonetheless, how to integrate process signals with physical information effectively remains a challenge.
(4) The hysteresis effect observed between internal weld defects and their corresponding locations (Ma et al., 2022c), as depicted in Fig. 18, poses challenges in accurately identifying and localizing internal and subtle welding defects, such as pores and cracks. Consequently, the establishment of a model integrating distance compensation and prediction, taking into account heat and mass transfer, could provide a pragmatic resolution.

5. Challenges and future opportunities

Considering the challenges discussed in each task in ML-based IPM-LDPW, we have identified some important directions for future research that could have a wider impact on this field.

5.1. Model interpretability

The issue of model interpretability is crucial in industrial applications. In classic machine learning, experts derive features from domain knowledge acquired through studies on mechanism elucidation and physical process modeling. However, deep learning is different, it treats feature extraction as a black box, thus generating high-dimensional features through deep learning that may be inscrutable to humans, yet prove to be highly effective. This raises the question of how we can understand the factors contributing to outputs and find optimization advice for welding process.

Physics-informed machine learning has demonstrated the potential in integrate data mathematical physics models seamlessly with good interpretability. Research has shown that transformations between the features of the latent space obtained by ML and the physically interpretable observables, will bolster the interpretability of the ML model.

5.2. Research shifting: macro to micro

The research focus for ML-based IPM-LDPW is recommend to shift from a macroscopic to a microscopic perspective. This shift is based on the fact that the solidification law of the molten pool is relatively consistent, which provides an opportunity to infer the microstructure of the weld seam by probing the inner temperature distribution of the weld pool. The main question is whether it is possible to reconstruct the thermal profile of the molten pool by the surface temperature distribution of weld pool. According to existing literature, using sensors that focus on the entire cycle of weld formation and **generative models combined with physical principles** is a possible solution.

Table 8
Recently typical methods of ML-based IPM-LDPW in the task of defects inspection.

Outputs	Sensors	Process Signals	Methods		Performance	Ref.
			Feature extraction	ML algorithms		
Porosity	Spectrometer (Ocean optics HR4000)	OE (400–800 nm)	LLE-WPT	ANN (ELM)	Acc: 92.97 %	(Yiming Huang et al., 2020b)
		OE (640–690 nm)			Acc: 96%	(Yiming Huang et al., 2020a)
Pores identification	High-speed camera (DMK 33UX174 monochrome)	Weld-pool image (640 × 480 pixel)	–	CNN	Acc: 96.1%	(B. Zhang et al., 2020)
Pores identification	Spectrometer (AvaSpec-ULS2048-8-USB2)	OE (318–420 nm)	Average and standard deviation of every 15 data	SVM (48-d)/ANN (16-d)	Acc: 98.08%/ 96.15%	Chen et al. (2019)
Pores and cracks identification	AE system (Kistler 8152B111)	AE	Curve statistical characteristics and flourier transform	Logistic regression/ANN	MSE: 1.72973/ 1.702703	Gaja and Liou (2018)
Pores identification	Si, Ge and InGaAs sensors and contact sensor PICO (Physical Acoustics)	OE (450–850 nm, 1250–1700 nm, 1000-1070–1200 nm), and AE (500–1850 kHz)	M-Band Wavelets	LapSVM	Acc: 89.1%	Shevchik et al. (2019)
Pore formation	Ge photodiode with a narrow band-pass filter (FB1070-10, Thorlabs Inc., USA), and piezo sensor PICO HF-1.2 (Physical Acoustics, Germany)	OE (1070 ± 2 nm), and AE (not mentioned)	WPT	Temporal CNN	Acc: 87%	Shevchik et al. (2020)
Porosity	IPG-LDD-700 coherent optical system, and a paraxial high-speed camera (phantom v611)	Keyhole image and keyhole depth (OCT)	EEMD and PCA	Feedback-GA-ANN	RRMSE: 0.0996/RMAE: 0.2459	Ma et al. (2022c)
Porosity	OCT	Maximum temperature on the bottom wall of the keyhole (numerical simulation) and keyhole depth (OCT)	SAE	Tandem two BDN	RRMSE: 0.0574/RMAE: 0.0953	Ma et al. (2022d)
Spatters	Spectrometer and photodiode	OE (350–750 nm, 1030 nm, 400–900 nm)	WPD-PCA	ANN	MSE: 0.015	You et al. (2015)
Welding type (Type 2 and 3 has the possibility of cracks formation)	PAC model UT-1000 AE sensor	High-pass-filtered AE signals (300–500 kHz, 200–300 kHz, 100–200 kHz)	RMS	ANN	–	Lee et al. (2014)
cracking susceptibility index	–	Process parameters and alloy properties	Solidification stress, ratio of vulnerable and relaxation times, solidification morphology, and cooling rate	SVM, DT, logistic regression, and linear regression	Acc: 85.3%/ 90.2%/70%/ 84.3%	Mondal et al. (2022)

'Acc' represents accuracy, 'MSE' represents mean squared error, 'RRMSE' represents relative root mean square error,' RMAE' represents relative max absolute error, 'RMS' represents root mean square, 'R2' represents determination coefficient.

5.3. Few/zero learning and transfer learning

The cost of data labeling in ML-based IPM-LDPW has been a constraint while training an ML model. It is quite challenging to achieve good generalizability with small-scale data. However, incorporating techniques from the **zero-shot/few-shot learning** literature can help overcome this challenge. Additionally, since the fusion welding process involves similar physical mechanisms, **transfer learning** can be applied to utilizing the data obtained from different types of welding technology by eliminating the distribution gap.

5.4. Open-source database

It is important to establish an open-source database for ML-based IPM-LDPW to build reliable and robust models. However, labeling data in the welding field is often expensive and requires manual labor. Creating a massive database is even more financially burdensome. The actual industrial scenarios are diverse, which creates a distribution gap with the scenarios presented by the limited experimental conditions of laser welding. Moreover, the lack of types and quantities of testing samples cannot fully verify the proposed model's reliability, which hinders the model's generalization ability. Therefore, only a freely shared database can provide more opportunities for developing welding-oriented LLM.

5.5. Model deployment

It is challenging to reduce the demand for computing resources while maintaining model performance when deploying deep learning models in real-world production due to the limited capacity of edge computing devices. As mentioned in the question, there are currently two approaches to addressing this problem. The first is the development of lightweight models that can deliver excellent performance. One possible research direction for this is knowledge distillation. The second approach involves the development of a software framework or a low-cost hardware that has high computing power.

6. Conclusion

This survey provides a comprehensive overview of modern research on machine learning-based IPM-LDPW systems. It covers the various physical mechanisms, sensing techniques, methodologies, advantages, and limitations of these systems, including classic and recent studies. The paper systematically introduces and analyzes representative in-process monitoring tasks in laser deep penetration welding that rely on classic machine learning and deep learning, including mechanical performance prediction, weld penetration estimation, and weld defects detection. At last, the unsolved challenges are outlined and the potential research directions are listed to address them.

Table 9
Common methods of weld defect detection.

Categories	Features & Advantages	Weaknesses & challenges
Hand-crafted features + classic machine learning	Mature theories and models can be used as reference;	Relies on manual experience or domain knowledge;
	Physical mechanism-based feature engineering can provide the interpretability;	Feature extractions based on traditional image processing and spectrum calculation are time-consuming;
	Good robust.	Prediction accuracy depends on the size and scope of the data used for training.
Deep learning-based methods	Modeling with raw data, no need for hand-crafted features;	Demands for massive scale labeled data;
	Can extract features that are unable to feature engineering based on prior knowledge;	Lack interpretability;
	Better model performance.	High computational resource requirements; Easy to overfit.
Hand-crafted features + Deep learning	Mature theories and models can be used as reference;	Relies on manual experience or domain knowledge;
	Physical mechanism-based feature engineering can provide the interpretability;	Feature extractions based on traditional image processing and spectrum calculation are time-consuming;
	Good robust.	High computational resource requirements; Demands for relatively small scale labeled data.
Deep learning-based features + Deep learning	Modeling with raw data, no need for hand-crafted features;	Demands for massive scale labeled data;
	Can extract features that are unable to feature engineering based on prior knowledge;	Lack interpretability;
	Better model performance.	High computational resource requirements; Hard to train due to error accumulation during two step training; Easy to overfit.
Physical informed machine learning	Good interpretability;	Relies on domain knowledge;
	Small data.	Poor scaling to more complex problem;
	Good robust; Good generalizability.	Poor convergence.

CRediT authorship contribution statement

Rundong Lu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ming Lou:** Investigation, Project administration, Writing – original draft, Writing – review & editing, Funding acquisition, Supervision. **Yujun Xia:** Methodology, Writing – original draft, Writing – review & editing, Supervision, Validation, Funding acquisition. **Shuang Huang:** Investigation, Software. **Zhuoran Li:** Investigation. **Tianle Lyu:** Investigation. **Yidi Wu:** Investigation. **Yongbing Li:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

All authors disclosed no relevant relationships.

Data availability

No data was used for the research described in the article.

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